

Outside Options and Labor Supply: Evidence from the Gig Economy*

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Abstract

We use randomized pay experiments among Uber drivers, paired with a natural experiment in access to a competitor, to examine how outside options shape labor supply to the firm. When hours are flexible, the firm-specific labor supply elasticity combines a market-hours component and a firm-substitution component. Access to a single competing platform nearly doubles drivers' firm-specific elasticity and cuts the implied monopsony markdown from 68% to about 50%. The same experiments identify sex differences: women are about twice as elastic to the market as men, yet no less elastic to their employer.

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1 Introduction

How much an employer can pay workers below their marginal product depends on where else those workers can go (Robinson, 1933; Manning, 2003). Models of imperfect competition predict that firms will pay lower wages to workers who are less elastic to the firm. A central tenet of this literature is that a worker’s outside options are a determinant of her firm-specific labor supply: workers with more employment opportunities are more elastic to the firm.

Estimating labor supply elasticities is challenging because it requires identifying exogenous variation in wages, either at the level of the market (to estimate market-level elasticities) or at the level of the firm (to estimate firm-level elasticities). Estimating firm-specific elasticities is particularly challenging as a researcher must ensure that the relevant variation is exogenous and firm-specific and that the firm does not adjust along other margins such as recruitment intensity or standards, job requirements or tasks, or non-wage amenities (Caldwell, Dube, and Naidu, Forthcoming). These requirements are hard to satisfy, even in experimental contexts. Linking labor supply elasticities to workers’ outside options is harder still: this requires exogenous variation in both wages and the outside employment opportunities faced by otherwise identical workers at the same firm.

We overcome these challenges by combining experimental wage variation, from a series of experiments we conducted among Houston Uber drivers, with quasi-experimental variation in access to an outside employer: Lyft. The ride-share setting is ideal for these questions: workers complete the same tasks, and recruiting standards and non-wage amenities are fixed. Drivers’ labor supply is also not constrained by firms’ hours choices.

We generated exogenous wage variation by offering random samples of drivers the opportunity to drive for one week with 20–50% higher earnings per trip. Both the week and generosity of the offer varied from driver to driver. We use quasi-experimental variation in access to Lyft—driven by the temporary withdrawal of Lyft from the Houston market—to identify the link between workers’ outside options and their labor supply to the firm. We refer to drivers with access to Lyft as “shifters” and those without such access as “non-

shifters.”

In the first part of the paper we present a model in which workers choose both how much to work (participation and hours) and for whom to work (firm substitution). We show that, when hours are flexible, the optimal monopsony wage markdown—the gap between marginal revenue product and the wage—depends on two elasticities: the total market-hours elasticity, which governs how a worker’s total hours—both whether she works and how much—respond to the wage, and the firm-substitution elasticity, which governs how she reallocates hours across firms. Existing monopsony research focuses on firm substitution and abstracts from the hours response; most labor supply research does the reverse.

Our first experiment, conducted while Lyft was absent from the Houston market, identifies the elasticity to the ride-share market: with no competing platform, a driver responds to a higher wage through her total ride-share hours, not by reallocating across ride-share platforms. We estimate an elasticity of one-half to nearly two-thirds—a 10% increase in the net-of-commission rate raises ride-share hours by about 5–6% (0.47–0.64), with a participation elasticity of 0.19. Elasticities are somewhat smaller for drivers who typically work more.

We oversampled female drivers so we could test whether men and women differ in their labor supply elasticities. The first experiment identifies this difference at the market level: women are about twice as elastic to the ride-share market as men (0.81 versus 0.42). The gap is not driven by other differences between men and women. No observable characteristic—usual hours worked, usual times of day worked, age, experience, or vehicle year—can explain the gaps in elasticities between male and female drivers. Because men and women here perform an identical task under the same pay algorithm and customer pool, the comparison is not confounded by differences in task assignment.

We then leverage quasi-experimental variation in access to an outside firm to examine whether changes in workers’ outside options affect how elastic they are to the firm. In the initial experiment, there is a single firm, so the firm-specific and market labor supply elasticities coincide. The Uber hours of drivers who can also drive for Lyft are roughly twice

as responsive to the Uber wage as those of drivers who cannot (firm-specific elasticities of 0.87 versus 0.47). Driver utilization rates—the share of online time spent en route to a pickup or on a trip—are also higher for these drivers, consistent with multi-apping across Uber and Lyft. Because these two groups come from experiments run at different times (before and after the re-entry of Lyft to the Houston market), we corroborate the contrast with an independent design that leverages experimental variation in promotional incentives and within-city differences in access to Lyft. These data produce the same pattern, though imprecisely estimated.

These elasticities speak directly to firms’ wage-setting power in the ride-share market. Under our framework the implied monopsony markdown falls from roughly 68% to 50%–54% with the addition of a new outside employer. In levels, a driver facing a single platform is paid only about a third of her marginal revenue product; access to one competitor raises this to roughly one-half—a 45–60% increase in pay, closing roughly a quarter of the gap to the competitive wage. We can also translate the elasticity gap into a cross-platform substitution elasticity, which measures how readily workers reallocate hours to the inside firm as its wage changes. Because this translation requires an assumption about Uber’s market share that our experiments do not pin down, we use a range of market-share values from the literature. Under our preferred specification, the firm-substitution elasticity is one and a half to three. Our estimates are roughly in line with quasi-experimental estimates of quit elasticities from the literature (see reviews in Caldwell, Dube, and Naidu, Forthcoming; Sokolova and Sorensen, 2021).

Robinson’s 1933 account of the gender wage gap turns on one requirement: that women supply labor less elastically to the firm. They do not. With access to an outside option, women’s firm-specific elasticity is 1.10, against 1.00 for men, so the implied markdown on women (48%) is, if anything, smaller than on men (50%). The market-level gap points the same way and is larger—women are twice as elastic to the market as men. Under the standard markdown, these elasticities imply that monopsony would pay women no less than men, and plausibly more: the channel explains essentially none of the gender gap in

this setting.¹

Our results speak to several literatures. First, to the literature on monopsony in labor markets. In a broad class of models, the firm-specific labor supply elasticity is informative about a firm’s wage-setting power. Our conceptual framework shows how the standard monopsony markdown extends to a flexible-hours setting, where the firm-specific elasticity combines a market-hours and a firm-substitution component. We use randomized wage variation, together with variation in workers’ access to a competing employer, to test the mechanism at the heart of these models: that better outside options make workers more elastic to the firm and so lower the markdown. We also provide one of the few experimental estimates of the firm-specific labor supply elasticity, and among the first broken down by sex. Our results thus contribute to a large literature that estimates these elasticities using quasi-experimental and structural approaches (Lamadon, Mogstad, and Setzler, 2022; Berger, Herkenhoff, and Mongey, 2022; Sharma, 2023).² Estimates in this literature range widely—from near 0.1 on online task platforms (Dube et al., 2020) to several times that in markets with close competitors. Our mechanism organizes this dispersion: firm-specific elasticities are high precisely where workers’ outside options are close and relatively undifferentiated, as our within-market variation in access to Lyft shows directly.

Second, our results contribute to a large literature on market-level labor supply elasticities. A concern in the firm-specific labor supply literature is that employment responses may be contaminated by changes in firms’ recruiting standards; a related concern in the market labor-supply literature is that observed labor supply responses reflect both workers’ and firms’ preferences: if firms offer fixed-hours packages, workers may not be able to choose their hours freely (Dickens, Lundberg, et al., 1993). Several studies use flexible-hours settings, where firms do not constrain workers’ hours, to study labor supply (Oettinger, 1999; Fehr and Goette, 2007; Farber, 2005; Stafford, 2015). However, the populations in these studies were predominantly male. We provide estimates from a similar environment, broken down by sex.

¹Cook et al. (2021) document a gender wage gap among Uber drivers.

²Caldwell, Dube, and Naidu (Forthcoming) provide a summary and classification of the recent literature.

Finally, our results contribute to a growing literature on ride-share labor markets (e.g., Athey, Camilo Castillo, and Chandar, 2025; Hall, Horton, and Knoepfle, 2023) and on gender gaps in ride-share (Cook et al., 2021). Our experiments build on Angrist, Caldwell, and Hall (2021) and complement Chen et al. (2025) and Chen et al. (2019), who use higher-frequency wage variation to study reservation wages and the value of flexibility. We instead ask how outside options shape labor supply to the firm.

The rest of the paper proceeds as follows. Section 2 develops the framework and the markdown decomposition; Section 3 describes the experiments and data; Section 4 presents market-level labor supply, overall and by sex; Section 5 turns to outside options—their effect on the firm-specific elasticity, the implied markdown, and sex differences; and Section 6 concludes.

2 Conceptual Framework

In a broad class of models, a firm’s market power in the labor market depends on the elasticity of its labor supply (Caldwell, Dube, and Naidu, Forthcoming). In this section, we show that workers with better outside options respond more to wage changes, raising this elasticity and limiting what the firm can extract. In flexible-hours environments, the relevant elasticity depends on both a participation margin—how many workers show up—and an hours margin—how much each one works. The addition of an outside employer can amplify both margins.

2.1 Firm-Specific Labor Supply

The textbook monopsony model considers a wage-posting firm that hires $L(w)$ units of labor at wage w , producing $Y(L(w))$. The firm sets w to maximize profits:

$$\max_w Y(L(w)) - wL(w).$$

The firm's optimal wage is

$$w^* = \frac{Y'(L(w^*))}{1 + 1/\tau(w^*)}, \quad (1)$$

where $\tau(w) = \frac{d \log L(w)}{d \log w}$ is the elasticity of labor supply to the firm. This equation provides the standard monopsony markdown: a higher elasticity means less market power and wages closer to marginal product (Manning, 2003).

The elasticity depends on the distribution of workers' reservation wages.³ Suppose the firm faces a continuum of potential workers of mass N . Each worker i has a reservation wage b_i —the lowest wage at which she is willing to work for the firm rather than take her best alternative, whether another employer, home production, or leisure. At wage w , the mass of workers who work at the firm is $N(w) = N \cdot F(w)$, where F is the CDF of reservation wages. When hours are fixed, $L(w) = N(w)$ and $\tau = \eta^{\text{firm}}$, where

$$\eta^{\text{firm}}(w) = \frac{f(w) \cdot w}{F(w)}.$$

When the density $f(w)$ is high, many workers are near indifference and a small wage cut causes many of them to leave the firm: labor supply is elastic. By contrast, when $f(w)$ is low, few workers are at the margin and a wage cut causes few workers to leave: labor supply is inelastic.

2.2 The Hours Margin

Now consider an environment where hours are flexible. In this case, a firm that raises its wage can see an increase in labor supply from two sources: new workers who enter as the wage rises, and workers already at the firm who raise their hours. Most prior work on monopsony focuses on the first margin and ignores the second. Suppose each worker i has a strictly convex cost of hours $v_i(h)$, with $v_i(0) = 0$, and pays a fixed participation

³In a broad class of models, the firm-specific elasticity reflects the firm's labor market power (Caldwell, Dube, and Naidu, Forthcoming). See the meta-analysis in Sokolova and Sorensen (2021).

cost $\chi_i > 0$ if she works.⁴ Conditional on working, hours satisfy $w = v'_i(h_i^*(w))$. Worker i works at the firm whenever the utility from doing so exceeds her participation cost χ_i :

$$wh_i^*(w) - v_i(h_i^*(w)) \geq \chi_i.$$

Her reservation wage b_i is the smallest w at which this inequality holds.

The optimal monopsony markdown still depends on the labor supply elasticity; this now includes an hours component. To express the total elasticity in additive form, suppose workers have a common cost of effort $v(h)$ and differ only in their fixed participation costs, with $\chi_i \sim G(\cdot)$. Let $u(w) = wh^*(w) - v(h^*(w))$ denote a worker's utility from working at wage w . By the envelope theorem, $u'(w) = h^*(w) > 0$, so a higher wage raises utility and weakly raises employment. The total number of workers at the firm is $N(w) = N \cdot G(u(w))$ and total labor supply is $L(w) = N(w) \cdot h^*(w)$. The total elasticity then decomposes:

$$\tau = \frac{d \log L}{d \log w} = \frac{d \log N(w)}{d \log w} + \frac{d \log h^*(w)}{d \log w} = \eta^{\text{firm}} + \varepsilon. \quad (2)$$

In this equation $\eta^{\text{firm}} = w g(u(w)) h^*(w) / G(u(w))$ is the firm-specific participation elasticity and $\varepsilon = wh^{*'}(w) / h^*(w)$ is the market-hours elasticity. In fixed-hours environments, η^{firm} is the relevant elasticity for calculating the firm's optimal markdown; in flexible-hours environments, the hours response also matters.

2.3 Outside Options and Labor Supply

The reservation wages in Sections 2.1 and 2.2 reflect whatever outside options workers already have—other employers, home production, leisure. But one aim of many policies is to either increase the number of employers (reduce concentration) or make it easier for workers to switch between existing employers. Suppose a new outside employer enters the

⁴The participation cost χ_i captures the value of the worker's best alternative to the firm—another employer, home production, or leisure—together with any fixed startup effort (activating the Uber app, accepting dispatches, and managing the platform). Allowing positive χ_i ensures that the marginal entrant chooses positive hours, so the extensive and intensive margins are well defined.

market, offering worker i earnings $S_i(r)$ for r hours, where S_i is increasing and strictly concave with $S_i(0) = 0$.⁵ Worker i now chooses inside hours h and outside hours r to maximize

$$wh + S_i(r) - v_i(h + r).$$

Effort depends on total hours $h + r$.⁶

The outside employer reshapes labor supply on both margins of Section 2.2: it changes who works for the firm and how many hours those who stay supply.

The Extensive Margin. Some workers find the new employer attractive and set $r_i^* > 0$. For those who keep positive hours at the inside firm, the marginal return to outside hours equals the inside wage, $S_i'(r_i^*) = w$; a worker who works only for the outside employer instead sets its marginal return equal to her marginal cost of hours, $S_i'(r_i^*) = v_i'(r_i^*)$. Others do not find the new employer attractive and set $r_i^* = 0$. For workers who take up the outside option, the inside firm must compensate them not just for the cost of an additional hour, but for the outside earnings they forgo.⁷

Reservation wages rise weakly for everyone, and strictly for any worker whose outside option has value on its own ($U_i^O > 0$)—even one who sets $r_i^* = 0$. Let F_0 denote the CDF of reservation wages without the outside firm and F_1 the CDF with it. Because each worker's reservation wage rises weakly, $F_1(b) \leq F_0(b)$ for all b : the distribution of reservation wages shifts in the sense of first-order stochastic dominance (FOSD). This leads to Result 1:

⁵Concavity of S_i arises naturally when the worker faces multiple outside employers with different wages. If worker i has access to jobs paying $w_1 > w_2 > \dots$, each for limited hours, she takes the best-paid work first—equivalently, accepts whichever platform offers the higher fare on each trip. The resulting $S_i(r)$ is piecewise linear and concave.

⁶Assuming effort depends only on total hours means that the cost of switching between platforms is only a function of total hours. Result 2 is robust to fixed costs of multi-apping, such as the \$5–\$100 per year that drivers pay for third-party switching apps (Section 3.1). Additive separability would not hold, however, if the marginal cost of reallocation rose in r independently of total hours. Then drivers would reallocate less than under frictionless switching and our estimates of $|\sigma|$ would understate the frictionless cross-platform response.

⁷Appendix B.1 shows that the participation threshold rises from b_{i0} (the smallest wage at which the inside utility $U_i^A(w) = \max_h [wh - v_i(h)]$ covers the fixed cost χ_i) to a new threshold $b_{i1} \geq b_{i0}$, where b_{i1} is determined by the inside utility net of the outside-only utility $U_i^O = \max_r [S_i(r) - v_i(r)]$.

Result 1. *A new outside employer leads to a first-order stochastically dominant shift in the distribution of reservation wages. If the pair of densities satisfies the monotone likelihood ratio property (MLRP, i.e., $f_1(b)/f_0(b)$ nondecreasing in b), $\eta^{\text{firm}}(w)$ rises at every wage w at which the firm hires a positive measure of workers and the new reservation-wage density is positive ($F_1(w) > 0$ and $f_1(w) > 0$), strictly so whenever the likelihood ratio is not constant below w . See Appendix B.1 for the proof.*

MLRP is plausible when the outside employer is available to everyone and offers everyone the same wage. In that case, workers with the weakest existing outside options—those with the lowest reservation wages—gain the most from a new alternative, so their reservation wages rise the most. This compresses the reservation-wage distribution from below, raising its floor. Because the firm-specific elasticity reflects how many workers sit near the margin, such compression tends to leave more workers marginal at a given wage, making labor supply more elastic. However, compression from below does not guarantee MLRP; Appendix B.1 provides the formal conditions.

Figure 1 illustrates Result 1 using an example that satisfies the MLRP condition. The red line plots the baseline density f_0 , which we assume is uniform for simplicity. Adding another employment option weakly increases all workers' outside options, yielding the blue density f_1 . Workers with the weakest existing outside options gain the most from the new employer, raising the floor of the reservation-wage distribution.

Panel B translates these densities into labor supply curves. Each worker participates if $b_i \leq w$, so labor supply at wage w is the CDF $F(w)$ scaled by the population mass; the inverse supply curve—wage as a function of labor supplied—is the inverse CDF. The red curve (f_0) shows a steep labor supply curve: a small wage cut causes only a modest drop in employment. The blue curve (f_1) shows a flatter labor supply curve: because more workers have high reservation wages, the same wage cut causes a larger employment loss. The firm faces a more elastic labor supply and sets a higher equilibrium wage.

The Intensive Margin. As before, with flexible hours, a wage increase raises a firm’s labor supply through more than one channel. Some additional hours come from new workers, drawn in as the wage clears their reservation wage—the participation response η^{firm} of Result 1. The rest come from workers already at the firm, who supply more hours as the wage rises—the market-hours response ε . When workers can hold more than one job, there is a third channel: a higher inside wage leads workers to shift hours they would have worked for the outside firm to the inside firm. The addition of an outside employer raises the elasticity to the inside firm through all three channels.

For a worker with positive hours at both firms, the first-order conditions give $w = S'_i(r)$: the inside wage equals the marginal return to outside hours.⁸ Define total hours $H := h + r$ and the inside share $\phi = h/H$.⁹ Since $h = H - r$, we can decompose the elasticity of inside hours to the inside wage as

$$\frac{\partial h}{\partial w} \frac{w}{h} = \frac{1}{\phi} \varepsilon - \frac{1 - \phi}{\phi} \sigma, \quad (3)$$

where ε is the market-hours elasticity (the elasticity of total hours H) and σ is the elasticity of outside hours to the inside wage. This captures the extent to which workers “leave” the outside firm in response to the change in the inside firm’s wage. Because $w = v'_i(H)$ regardless of the outside firm, total hours H and hence ε are unchanged by the presence of the outside employer. Strict concavity of S_i implies $\sigma < 0$: a higher inside wage reduces outside hours.¹⁰ In the empirical sections and tables, we report the magnitude $|\sigma|$ for ease of interpretation.

We use this expression in our empirical exercises. Without an outside firm (i.e., when Lyft is out of the market), $\phi = 1$ and the inside-hours elasticity equals ε . With an additional outside employer (here, Lyft), each worker who supplies positive outside hours has an

⁸Appendix B presents a similar model in which workers can work for a single firm at a time.

⁹We do not observe individual drivers’ inside shares; in the calibrations below we proxy ϕ with aggregate measures of Uber’s share of ride-share activity, which coincide with the individual share when drivers are homogeneous.

¹⁰Apply the implicit function theorem to $S'_i(r) - w = 0$. Since $S''_i < 0$, $\partial r / \partial w = 1 / S''_i < 0$.

elasticity above the single-firm value by $(1 - \phi)/\phi \cdot (\varepsilon - \sigma)$ —a wage increase both raises total hours and reallocates hours from the outside firm. The amplification grows as the inside share ϕ falls.

Result 2. *A new outside employer weakly raises participating workers’ inside-hours elasticity relative to the single-firm hours elasticity, with a strict increase for workers who allocate positive hours to the outside firm. If the densities satisfy MLRP and workers share a common market-hours elasticity ε ,¹¹ the total firm-specific elasticity τ rises at every wage—exactly when hours conditional on working are homogeneous across workers, and as an approximation otherwise. See Appendix B.2 for the proof.*

When there is a single firm, the total elasticity inclusive of hours is $\tau = \eta^{\text{firm}} + \varepsilon$ (equation 2). When workers can work for multiple firms at the same time and choose these hours freely, it generalizes to

$$\tau = \eta^{\text{firm}} + \frac{1}{\phi}\varepsilon - \frac{(1 - \phi)}{\phi}\sigma. \quad (4)$$

If both types of workers share the same η^{firm} and ε , the difference between the two groups isolates the intensive-margin amplification $(1 - \phi)/\phi \cdot (\varepsilon - \sigma)$, which combines the $1/\phi$ scaling of the market-hours response with the reallocation channel governed by the firm-substitution elasticity σ . This decomposition holds exactly for each worker i , with worker-specific ϕ_i and σ_i .

2.4 Mapping the Framework to Data

Our experiments change the inside wage w for a random subset of drivers. Non-shifters—drivers without access to Lyft—correspond to the single-firm case: they have no outside employer, so the outside-earnings function $S_i \equiv 0$ and they allocate all their hours to Uber ($\phi = 1$).¹² Their total response therefore identifies $\tau_{\text{non}} = \eta^{\text{firm}} + \varepsilon$ (equation 2).

¹¹This requires v_i to take the isoelastic form $v_i(H) = \alpha_i H^{1+1/\varepsilon}/(1+1/\varepsilon)$, where α_i varies across workers but the exponent is common. Workers differ in hours ($H_i = (w/\alpha_i)^\varepsilon$) but share the same elasticity.

¹²This is a Lyft-only normalization, not a literal claim that non-shifters have no other work options: S_i is the Lyft earnings function and ϕ is Uber’s share of a driver’s ride-share hours. As we discuss at the end of

We write $\bar{\varepsilon} := \eta^{\text{firm}} + \varepsilon$ for this participation-inclusive total market-hours elasticity. For drivers with no ride-share alternative, the firm-specific and market-level total elasticities coincide. Shifters—drivers who can also work for Lyft—correspond to the multi-firm case (equation 4); their response reflects both ε and the firm-substitution elasticity.

The key identifying assumption is that shifter status is exogenous to labor supply preferences. Vehicle-age eligibility or Lyft’s temporary exit from Houston determines access to Lyft; neither reflects driver preferences.¹³ The cross-group comparison further requires that, absent access to Lyft, shifters and non-shifters have the same labor supply preferences: the same market-hours elasticity ε and the same baseline distribution of attachment to Uber.

Under these assumptions, the gap in total elasticities between shifters and non-shifters decomposes into the two channels of the framework:

$$\tau_{\text{shift}} - \tau_{\text{non}} = \underbrace{(\eta_{\text{shift}}^{\text{firm}} - \eta_{\text{non}}^{\text{firm}})}_{\text{Result 1: participation channel}} + \underbrace{\frac{1 - \phi}{\phi}(\varepsilon - \sigma)}_{\text{Result 2: intensive-margin channel}}.$$

The first term is the observed extensive-margin effect of Lyft access on participation (Result 1, the empirical gap reported below). The second is the intensive-margin reallocation channel from Result 2. Empirically, we estimate τ_{shift} and τ_{non} directly. To recover the firm-substitution elasticity σ we use the intensive-margin decomposition (equation 3), which relates the shifter and non-shifter log-hours elasticities given a calibrated inside market share ϕ (Section 5.1); we report the magnitude $|\sigma|$ in the empirical results. Because ϕ is not identified by our design, we report $|\sigma|$ across a range of values from the literature. Table 1 summarizes the parameters and identifying variation.

While we model Lyft as the sole outside employer, drivers may also work for other platforms. Our identification strategy does not require that non-shifters lack all outside

this subsection, our identification strategy assumes that shifters and non-shifters substitute with employers other than Lyft in the same way, so that substitution with non-Lyft alternatives is common to both groups and cancels in the difference.

¹³The primary concern is that vehicle-age eligibility may be related to underlying wealth—and therefore reservation wages. To address this concern we perform a number of additional specification checks, detailed in Section 5.1. In particular, we show that vehicle age does not predict elasticities in Houston, and that available proxies for wealth do not predict elasticities.

options—only that shifters and non-shifters substitute with employers other than Lyft in the same way. Any firm substitution with non-Lyft employers is then common to both groups and cancels in the difference.

Like most experiments, ours vary pay only temporarily, so we estimate short-run elasticities. These may differ from the long-run elasticities relevant to a firm’s wage-setting: income effects make the long-run elasticity smaller than the short-run one, while adjustment costs make it larger (Bassier and Manning, 2025). The harder problem is not the horizon. What determines a firm’s market power is how labor supply responds when workers’ outside options change, and recovering that takes variation in market structure, not in wages alone. Most experiments hold the outside option fixed; ours varies it, through Lyft’s entry and exit. Identifying how outside options shape the long-run elasticity would require both at once—exogenous variation in market structure sustained over a long horizon—which is harder still.

3 Empirical Setting and Data

We identify labor supply elasticities using data from a series of field experiments at Uber, as well as data from an Uber-run promotion in which Uber randomized incentives.

3.1 Background on Ride-Share

Uber is a global Transportation Network Company (TNC) that connects drivers and riders through a mobile app. Uber launched peer-to-peer operations in mid-2012 and by 2019 had over 900,000 active drivers in the United States. Most U.S. cities impose few barriers to becoming a ride-share driver.¹⁴

Uber drivers choose when and where to work within Uber’s service region. During our experiments, drivers earned per mile and per minute for each trip completed. These rates

¹⁴The most prominent counterexample is New York City, where ride-share drivers must obtain licenses from the Taxi & Limousine Commission. We do not use any data from New York City.

rose at certain times of day and in certain locations through Surge pricing. For many years, Uber drivers paid a fixed fraction of their trip receipts to Uber as the “Uber fee.”¹⁵ This fee varied across drivers based on their city and when they joined the platform. Appendix Figure A1 shows an example trip statement.

Many ride-share drivers work for multiple platforms, a practice known as “multi-apping.” A 2021 ride-share blog survey estimated that two-thirds of drivers drove for both Uber and Lyft (Campbell, 2022). At the time of our experiments, these two companies dominated the on-demand gig market (Campbell, 2017).

Drivers on multiple platforms may shift between them on a shift-to-shift basis, choosing whichever app offers the highest earnings. They may also multi-app at high frequency, keeping both apps on during downtime and accepting the first dispatch. Because drivers earn only when a passenger is in the car, this raises hourly earnings.

Low-frequency multi-apping is straightforward, but high-frequency multi-apping requires effort. Online forums (Appendix Figure A3, Panel A) show that drivers must actively toggle between applications, turning off one upon accepting a trip on the other. Drivers exchange information on multi-apping through word of mouth, online forums, and ride-share blogs (Appendix Figure A3, Panel B). Several companies (e.g., Mystro, Upshift, and QuickSwitch) sell third-party apps that help drivers navigate between the two interfaces. These apps—recommended by popular ride-share blogs—charge \$5/week, \$12/month, or \$100/year, suggesting that multi-apping is both profitable and costly (Appendix Figure A2).

Some ride-share drivers cannot drive for both Uber and Lyft because only one platform operates in their market. Between November 2016 and May 2017, Lyft did not operate in Houston.¹⁶ Even in cities where both platforms operate, some drivers are ineligible for one platform based on vehicle age. For many years, Lyft required Boston and San Francisco

¹⁵In 2017 Uber loosened the link between driver and rider pay. Riders and drivers now face distinct per-minute and per-mile rates.

¹⁶Houston’s decision to require ride-share companies to implement fingerprint background checks prompted Lyft’s exit. While both Uber and Lyft left the Austin market over a similar law, only Lyft left the Houston market. In May 2017 the Texas state legislature passed statewide legislation that allowed companies to use name-based background checks.

drivers to use cars with a model year of 2004 or newer, while Uber allowed older vehicles (Angrist, Caldwell, and Hall, 2021).¹⁷

3.2 Earnings Accelerator Experiments

We generated wage variation through randomized experiments. Variation in access to Lyft arose independently of the experiments, from Lyft’s temporary withdrawal from Houston and from vehicle eligibility requirements. Ride-share companies routinely run promotions that change driver pay without affecting rider prices. We modeled our experiments on these promotions and designed them with local city teams to resemble typical Uber promotions, advertising them as the “Earnings Accelerator.”

We conducted two Earnings Accelerator experiments in Houston. The first ran in spring 2017, when Lyft was temporarily out of the market; the second ran in fall 2017, after Lyft had returned. While Lyft was absent, Uber was the only major ride-share company operating in Houston.¹⁸ We use the first experiment—a pure non-shifter sample—to estimate the labor supply elasticity to ride-share. The contrast between the two experiments is our preferred quasi-experimental estimate of how adding an outside option affects firm-specific elasticities; because the experiments differ in timing as well as in Lyft access, we support it with within-market evidence (the Boston eligibility contrast and the Individual Driver Bonus (IDB) program) and the robustness checks below. Because the second experiment was cut short, some specifications also use data from a third Earnings Accelerator experiment in Boston (Angrist, Caldwell, and Hall, 2021). Appendix Table A1 compares the three Earnings Accelerator experiments. Appendix Figure A4 depicts the design of the first Houston experiment and Appendix C.2 provides more information on all experiments.

¹⁷Both Uber and Lyft had additional requirements for driving for their “premium” services.

¹⁸Ride-share drivers today have many gig alternatives at Amazon, at food delivery companies, and at smaller ride-share companies. This was not the case in Houston when we conducted the experiment in mid-2017, as contemporaneous ride-share surveys confirm (Campbell, 2017).

Sample Selection. We used similar sample-construction protocols and randomization designs in both Houston experiments. We first identified “eligible drivers” who satisfied two criteria: (1) they were active on the Uber platform (at least four trips in the past month), and (2) they averaged between 5 and 40 hours per week, conditional on driving, in the four weeks prior to the experiment. To ensure a mix of full-time and part-time drivers, we grouped drivers into bins by usual weekly hours and randomly selected subsets from each bin. The low group consisted of drivers averaging between 5 and 15 hours per week, the high group of drivers averaging between 15 and 25 hours per week, and the very high group of drivers averaging between 25 and 40 hours per week. Within each bin, we randomly selected drivers and oversampled women.¹⁹ We selected 2,020 drivers for the first Houston experiment and 2,100 for the second.

Treatment. We offered selected drivers the opportunity to earn a fixed percentage more on every trip for a week. Half of the drivers in each randomization stratum (defined by usual hours and sex) received the offer in week one; the other half received it the following week.²⁰ We refer to a driver as treated in a week when she is offered the higher pay and as a control in a week when she is not. In the first experiment the offer rotated across weeks; in the second, we have only a single week of comparable data. We told drivers this would result in an “X% higher payout.” In the non-shifter experiment, we implemented these increases as reductions in the Uber fee, so X was either 39% or 25%, depending on the commission a driver faced (a driver paying a 28% commission who keeps the full fare earns $1/0.72 - 1 \approx 39\%$ more; one paying 20% earns $1/0.80 - 1 = 25\%$ more). In the second experiment, we offered increases of between 20% and 50%, with no explicit mention of the Uber fee.²¹ In both cases, the offers were quoted in proportional terms

¹⁹The Boston Earnings Accelerator experiment was structured similarly. It did not have a “very high” hours bin and did not oversample female drivers.

²⁰We had planned to use this rotating design in both waves of the Houston experiment. However, Uber made unexpected changes to the app, which removed our ability to change drivers’ pay in a way that would be visible on a trip-by-trip basis after the first week of the second experiment.

²¹The fee was highly salient to drivers because it appeared on each trip receipt and weekly pay statement (Angrist, Caldwell, and Hall, 2021). In June 2017, Uber changed its pay policy to loosen the link between rider fares and driver earnings—known as “upfront pricing”—and removed the concept of the Uber fee. What

(e.g., 39% more or 25% more). While the treatment was active, drivers who accepted saw increased earnings reflected in-app after every trip.

The offers stated that the promotion would apply to all trips that week, including those with Surge pricing. Drivers received offers via e-mail, text message, and the Uber app. Figure 2 shows a sample e-mail and text message from the first Houston experiment. These messages (and the in-app notification) linked to Google Forms like those typically used in Uber promotions. The forms were pre-filled with a driver's unique Uber identifier and included promotion details and consent language. We sent offers one week before the promotion went live; drivers had one week to accept by clicking "yes" on the Google Form. Around two-thirds of drivers in each experiment accepted. The offer should have attracted all drivers, but Uber drivers receive many messages each week and some ignore them.²² Given our messaging and Uber's general approach to driver promotions, drivers had no reason to anticipate higher future wages from participating.

Market-Level Concerns. We designed each experiment to be small—roughly 2,000 drivers out of more than 15,000 active in Houston—so it would not affect equilibrium wages. Large changes in the number of drivers on the road could affect equilibrium wages through wait times and Surge pricing. In the second experiment, large supply shifts away from Lyft could also have prompted Lyft to adjust Surge pricing or promotions. Because the treated share was small, none of these channels should have produced market-wide effects.²³

Taxi Experiments. Some specifications in Section 4—which provides estimates of the labor supply elasticity to the ride-share market—also use data from "taxi" experiments con-

drivers earned per trip did not change; it remained a function of a base fare plus a per-mile rate and a per-minute rate. The Uber fee was, however, no longer a relevant pay concept. We took advantage of this change to generate more variation in pay in the second experiment. Appendix C provides more details.

²²Appendix Figure A6 shows that drivers who drove the week the offers were sent out were more likely to accept the offer, since one delivery channel was the Uber app, which drivers may not open if they do not drive. Among those who drove, there was no relationship between hours worked and acceptance.

²³The first stage is consistent with this: the realized increase in treated drivers' net-of-commission rate matches the offered fee reduction scaled by take-up (Table A2), with no sign of offsetting equilibrium adjustment in wages or wait times.

ducted among drivers who accepted the initial offer in our non-shifter experiment. These experiments give us more power for subgroup analysis. In these experiments we offered treated drivers additional weeks of higher earnings in exchange for an up-front payment, much like the lease a taxi driver pays a medallion holder. Although these offers attracted only drivers who intended to drive enough to pay off the lease, they generated additional wage variation at much lower cost. We label specifications that include these data “taxi” specifications, and refer to the initial treatment weeks—in which the treatment came at no up-front cost—as the “fee-free” driving weeks.

Appendix C provides detailed information on the implementation of each Earnings Accelerator experiment, including sample counts, treatment offers, and messaging. Table A3 confirms that randomization is balanced across all three experiments.

3.3 The Individual Driver Bonus Program

We supplement the Earnings Accelerator experiments with data from a separate Uber program: the “Individual Driver Bonus.” The IDB operated in San Francisco and offered drivers lump-sum bonuses for exceeding pre-specified trip thresholds during a given time period (weekday or weekend). These bonuses created upward notches in drivers’ budget sets at the assigned trip thresholds (Figure A8). Uber randomly assigned drivers to “high” and “low” bonus amounts within eight activity-based strata. Appendix E reports the bonus amounts and excluded region by trip threshold, and shows that, conditional on strata, high- and low-bonus drivers are statistically indistinguishable on observables.

This program provides a second, independent source of variation for testing whether outside options raise labor supply to the firm. The IDB differs from the Earnings Accelerator in three ways: it operates in San Francisco rather than Houston; it generates variation in trips rather than hours; and it uses notch-based bonuses for exceeding trip thresholds rather than increases in drivers’ hourly earnings. It has the advantage that Lyft access varies across drivers in the same city at the same time. This within-market variation complements the Earnings Accelerator, whose Houston identification comes from Lyft’s exit and re-entry

over time; the analogous cross-driver contrast is too imprecise in the smaller Boston sample (Angrist, Caldwell, and Hall, 2021). Within the IDB, we define “shifters” as drivers whose vehicles were new enough (model year 2004 or newer) to qualify for Lyft in San Francisco, and “non-shifters” as those confined to Uber. These were the vehicle-age requirements at the time we ran the experiment.

3.4 Data and Descriptive Statistics

Our Earnings Accelerator data include weekly hours, earnings, and utilization for all drivers in the experiment, along with a limited set of driver characteristics—age, sex, and zip code—and information on each driver’s car. Uber does not collect detailed demographic data on drivers.

Most of our analysis examines how drivers’ hours respond—on either the intensive or extensive margin—to experimental wage changes. Following the ride-share literature, we define a driver as working whenever the Uber app is on and the driver has indicated availability for dispatch (Angrist, Caldwell, and Hall, 2021; Cook et al., 2021). Our hours measure therefore includes (1) time waiting for a dispatch, (2) time traveling to a pickup, and (3) time on a trip. The appendix reports complementary results using gross earnings as the dependent variable.²⁴

Table 2 summarizes the four samples. The two Houston Earnings Accelerator experiments include 2,020 and 2,100 drivers, the Boston experiment 1,600, and the IDB 58,928 drivers in San Francisco. Because we oversampled women in the Houston experiments, women make up 52% and 39% of those samples, against 14% in Boston and 18% in the IDB. Drivers average between 9 and 16 months on the platform and, where measured, around 20 hours per week in the weeks before the experiment. Vehicle vintages are similar

²⁴Uber distinguishes between gross earnings—which include Uber’s own platform promotions but not our experimental incentives, which operated through the fee—and net earnings—which subtract the amount the driver paid in Uber fees. We use gross earnings to avoid the mechanical relationship between our promotion (which reduced the fee drivers paid) and drivers’ earnings. Neither gross nor net earnings deduct costs the driver may incur, including gas or vehicle depreciation. To the extent that driver costs are approximately proportional to hours driven, this substitution is without loss of generality.

across samples (model year ≈ 2013), except in Boston, where cars are older on average (2010)—consistent with the looser vehicle-age requirement there.

We sampled drivers within strata defined by hours and sex to obtain enough high-hours drivers and adequate precision for our subgroup analyses. As a result, our sample overrepresents high-hours and high-experience drivers and is not, without reweighting, representative of the Uber driver population. We present reweighting exercises below that examine whether our estimates generalize to the full pool of active drivers.

4 Market-Level Labor Supply

We use the Earnings Accelerator experiments to estimate the elasticity of ride-share labor supply to the wage—first overall (Section 4.1), then separately for male and female drivers (Section 4.2).

4.1 The Elasticity of Labor Supply to Ride-Share

We estimate labor supply elasticities by regressing labor supply outcomes for driver i in week t on the net-of-commission rate (analogous to a net-of-tax rate), on the strata used for randomization, and on baseline covariates.²⁵ We use the log net-of-commission rate as the endogenous variable, following the public-finance literature on taxable-income elasticities; unlike the wage, it is observed even for drivers who do not drive. Because the realized net-of-commission rate depends on whether drivers accepted our offer, we instrument for it using the randomly assigned treatment offers Z_{it} .²⁶ Formally, we estimate:

$$\begin{aligned} y_{it} &= e \log(1 - \kappa_{it}) + \beta X_{it} + \xi_{it}, \\ \log(1 - \kappa_{it}) &= \gamma Z_{it} + \lambda X_{it} + \nu_{it}. \end{aligned} \tag{5}$$

²⁵The baseline covariates include the number of months a driver has been on the Uber platform, an indicator for whether a driver uses Uber’s “vehicle solutions” leasing program, one lag of log earnings, and a constant.

²⁶Appendix Table A4 shows that OLS estimates of labor supply elasticities are biased downwards. See Appendix D for more details.

In the baseline over-identified model, Z_{it} is a vector of indicators, one for each week-by-offer combination, equal to one when individual i was assigned that offer in week t . The just-identified model collapses this to a single treatment indicator. We cluster standard errors at the driver level throughout.

The coefficient of interest is e , the elasticity of the outcome with respect to the net-of-commission rate. When y_{it} is log hours, e is the intensive-margin labor supply elasticity. When y_{it} is an indicator for positive hours, it is a semi-elasticity, and we recover the participation elasticity by dividing by the sample participation rate.

First-Stage Estimates. Table A2 provides information on driver opt-in decisions and on the first stage for the first Houston experiment. Columns 1–2 report that the average offer was a 0.29 log-point ($\approx 34\%$) increase in the net-of-commission rate ($\log(1 - \kappa_{it})$). Columns 3–4 show that 59–67% of treated drivers accepted the offer. Because not all drivers accepted, the realized increase was smaller: Columns 5–6 show a 0.17–0.20 log-point impact on treated drivers’ net-of-commission rates, close to the offered increase scaled by the take-up rate.

Two features of the first stage support our interpretation of the experiment as a clean wage shifter. First, the realized impact on hourly earnings (Columns 7–8, 0.16–0.17 log points) tracks most of the impact on the net-of-commission rate, confirming that the fee reduction largely passed through to take-home pay.²⁷ This near-complete pass-through is consistent with the absence of market-level spillovers. These spillovers are also unlikely given the size of our experiments.

Main Estimates. Column 1 of Table 3 shows an extensive-margin participation elasticity of approximately 0.19: a 10% increase in the net-of-commission rate raises the probability a driver works by roughly 2%. Appendix D.6 presents hours distributions for treated and untreated compliers, following the methodology of Abadie (2002). The extensive-margin

²⁷The impact on realized log hourly earnings is somewhat smaller than the impact on the net-of-commission rate—a statistically significant ($p < 0.001$) but economically modest shortfall of a few log points—suggesting that marginal hours were somewhat less productive. Appendix D provides further details.

effect appears as the drop in mass at zero hours; beyond it, treatment shifts the entire distribution of positive hours rightward.

Because treatment moves the extensive margin, the hours elasticity has to account for both intensive- and extensive-margin responses. Our preferred estimate comes from the IV-Poisson model (Columns 4–5 of Table 3), which uses the full sample, including drivers with zero hours; it requires only a correctly specified conditional mean and is robust to heteroskedasticity. The IV-Poisson estimates imply a total market-hours elasticity $\bar{\varepsilon}$ of 0.47–0.64: a 10% increase in the net-of-commission rate raises ride-share hours by roughly 5–6%. These are close to the log-hours 2SLS estimates, which condition on drivers with positive hours and so estimate a conditional-positive-hours elasticity (0.42–0.69, Columns 2–3) that we map to the framework’s intensive elasticity ε .²⁸

We take the IV-Poisson estimate directly as our measure of the total market-hours elasticity $\bar{\varepsilon} = \eta^{\text{firm}} + \varepsilon$ (equation 2)—the elasticity of total ride-share hours to the wage.²⁹ These elasticities are identified from drivers who respond to the offer—the margin on which a wage-setting firm’s markdown depends. Reweighting to the full driver population barely changes them (Appendix Table A22 compares compliers to the pool of experimental drivers; Appendix D.4 reweights to the driver population).

Robustness. The appendix shows that the market-hours elasticities are robust to using a just-identified model and to controlling only for randomization strata, separately for male and female drivers (Table A6), and to using fares collected as the dependent variable (Ta-

²⁸Because treatment also shifts participation, the log-hours estimate reflects a composition effect from treatment-induced entry in addition to the pure intensive response. The near-equality with the Poisson estimates suggests this composition effect is small. An early version of this paper used the inverse hyperbolic sine of hours; given recent evidence on the instability of those estimates (see Chen and Roth, 2024), we prefer the Poisson model.

²⁹The two empirical objects map to the framework as follows: the log-hours 2SLS estimates measure the intensive elasticity ε , which the framework holds common across groups; the IV-Poisson estimates measure the participation-inclusive total $\bar{\varepsilon}$. We use $\bar{\varepsilon}$ (and the analogous total elasticities for shifters) wherever total responses are required—the markdown calculations—and the intensive log-hours estimates when recovering the intensive-margin object $|\sigma|$ (Section 5.2), consistent with the decomposition in equation 3, which is stated in intensive elasticities. The additive identity $\bar{\varepsilon} = \eta^{\text{firm}} + \varepsilon$ holds exactly under the framework’s common-hours assumption (Section 2.2). More generally the two margins combine with hours weights.

ble A7). Appendix Table A5 reports specifications based on IV-Tobit models.

Interpretation. Because Lyft was absent from the Houston market and few other gig platforms were available, raising drivers’ net-of-commission rate at Uber raised the return to ride-share work as a whole. We therefore interpret these estimates as market-hours elasticities (ε and $\bar{\varepsilon}$)—the response of hours supplied to the ride-share market, holding the marginal utility of wealth constant—rather than the ride-share firm-specific elasticity τ , which also reflects substitution toward an outside option. The wage increases are temporary: the additional earnings from the two-week experiment (roughly \$200 on average) are small relative to annual income, and Appendix Table A8 shows that treatment did not affect labor supply in the weeks before or after the experiment. As discussed in Section 3, Lyft and Uber captured the vast majority of ride-share work in Houston at the time, and other gig work was limited; during this experiment, Lyft was temporarily absent from the market.

4.2 Sex Differences in Market-Level Labor Supply

We oversampled female drivers so we could estimate sex-specific elasticities. Here men and women face the same firm, the same randomized wage offers, and identical tasks within a single experiment. As a result, we isolate differences in wage responsiveness; in other settings differences in job content, occupational sorting, and timing can confound comparisons.

Table 4 reports market-hours elasticities by sex. Women are roughly twice as responsive as men: a 10% increase in the net-of-commission rate raises men’s ride-share hours by about 4% and women’s by about 8% (Column 5, Poisson), and we reject equality of the male and female elasticities in the Poisson specification. The male-female gap in elasticities appears on both the intensive and extensive margins. The same wage increase raises women’s probability of driving by about 2.5% but leaves men’s participation nearly unchanged (0.25 versus 0.11, Column 1). The female point estimate remains roughly twice the male estimate across a broad set of specification checks; we present results which use

a just-identified model, which control only for randomization strata, and which use fares collected instead of hours (Appendix Tables [A6](#) and [A7](#)).

Although men and women here do the same job, they differ on average along several dimensions—age, platform tenure, the hours and days they drive—that could themselves shape wage responsiveness. To probe whether the gap in elasticities reflects characteristics correlated with sex rather than sex itself, Appendix Table [A9](#) re-estimates market-hours elasticities by sex within subgroups defined by these characteristics. Elasticities vary little by age or platform tenure (Columns 1–4). In addition, though men and women work at different times of day, the gap is similar when we compare drivers who work late nights, weekends, or school-pickup hours (Columns 5–7), making it unlikely that differences in schedules or childcare explain the gap. We also see a similar gap within high-hours drivers (on the intensive margin), who have little scope for employment outside Uber—weighing against an explanation based on outside (non-gig) work. While female drivers have newer cars (suggesting perhaps greater wealth), we find no difference in elasticities by car year (unreported).³⁰

These comparisons hold driver characteristics fixed. Of course, men and women also differ on average. Because we stratified the sample to make them comparable on usual hours, our design likely understates the gap between men and women for the average Houston driver. For instance, elasticities are larger among low-hours drivers, and women typically drive fewer hours. When we reweight the sex-specific elasticities to the active-driver population, allowing for differences in both hours and experience, the gap widens rather than closes—from 0.44 in our experimental sample to 0.50 (the reweighted female elasticity rises from 0.85 to 0.99 and the male elasticity from 0.41 to 0.49; Appendix Table [A22](#) compares compliers to the pool of experimental drivers).

³⁰In unreported results we also proxied wealth with median income in a driver’s zip code and found little heterogeneity. Because the experiment ran while Lyft was absent, substitution to Lyft cannot drive the heterogeneity; and outside non-gig work would explain the gap only if women were more attached to outside employment than men, which is not consistent with the Houston descriptives in Appendix [D.5](#).

5 Outside Options and Labor Supply

We next examine how labor supply responses change when drivers can also work for a competing platform. We use quasi-experimental variation in access to Lyft to estimate the firm-specific elasticity, translate it into an implied wage markdown, and test whether it differs by sex.

5.1 The Effect of an Outside Firm

We estimate the same model, stacking data from the two Earnings Accelerator experiments. Consistent with Result 2, Table 5 shows that shifters (drivers with access to Lyft) are more elastic than non-shifters across all specifications. Our preferred estimates come from the IV-Poisson model using only Houston data (Column 5). Non-shifters have an elasticity of 0.47 and shifters have an elasticity of 0.87; the difference is marginally significant ($p = 0.051$) and economically large, with shifters roughly twice as wage-responsive. In response to a 10% increase in the Uber wage, a driver with access to Lyft increases her Uber hours by 8.7%, while a driver without such access increases hers by only 4.7%. The gap is meaningful across specifications, and pooling in the Boston Earnings Accelerator makes it more precise (Column 6).

The conceptual model in Section 2 attributes this additional responsiveness to two channels: access to an outside option draws more drivers to the participation margin (Result 1), and drivers reallocate hours they would have worked for Lyft to Uber when the Uber wage rises (Result 2).

A concern is that, because shifters and non-shifters are drawn from different experiments, they differ somewhat on baseline observable characteristics—sex, experience, and baseline hours (Appendix Table A10). The experimental first stages are, however, comparable: treated shifters were offered a 0.24 log-point increase in the net-of-commission rate and opted in at 65%, similar to the offer and take-up among non-shifters (Appendix Table A11).

As with the sex comparison, one concern is that the gaps we find reflect differences attributable to other observable characteristics—rather than differences attributable to access to Lyft. We address this with two reweighting exercises. First, we reweight non-shifters to match the shifter distribution of sex, platform tenure, baseline log earnings, and participation in Uber’s vehicle-solutions leasing program, using entropy balancing to achieve exact balance on these covariates (Hainmueller, 2012). The gap is essentially unchanged ($\tau_{\text{shift}} - \tau_{\text{non}} = 0.35$ using Houston data alone and 0.50 pooling Boston), and it is positive within every quartile of the estimated propensity score (Appendix Table A12). Second, we let elasticities vary by bins of baseline hours and experience and recompute the gap holding the covariate distribution fixed at the non-shifter mix. Figure A5 shows the gap is little changed by either adjustment.

Specification choices also do not appear to drive our results. Appendix Table A13 shows that the pooled shifter–non-shifter gap in hours worked remains positive and significant under both just-identified and strata-only models (the extensive-margin gap is smaller and imprecise in these specifications), and that the gap holds separately for men and women. We also perform permutation checks, reassigning shifter status across the pooled Houston sample 1,000 times and re-estimating the IV-Poisson gap; this yields a one-sided p -value below 0.001 (Appendix Figure A7). Finally, estimating the shifter and non-shifter elasticities from complier mean hours following Abadie (2002), rather than from the IV coefficients, yields a similar gap (Appendix Table A19).

Additional Evidence on Reallocation. Drivers’ utilization rates provide some additional evidence for the reallocation channel. Multi-apping raises a driver’s utilization—the fraction of online time spent en route to a pickup or carrying a passenger—because a driver running both apps can accept a trip from whichever platform dispatches first and so spends less time idle, waiting for a request. If shifters have a mix of Lyft and Uber hours, their Uber utilization should be higher. Appendix Table A14 shows that utilization rates are significantly higher for shifters than non-shifters (0.615 versus 0.586, $p < 0.001$), with a

similar gap for both sexes.³¹ This pattern is consistent with shifters multi-apping during idle time.

Additional Evidence: The Individual Driver Bonus Program. As noted, our evidence linking changes in outside options to changes in labor supply elasticities comes from comparing two experiments run in the same city at different times. To address the concern that the pre-post comparison reflects time effects, we gathered data from an Uber-run promotion in a different city. As described in Section 3.3, the program offered drivers lump-sum bonuses for exceeding pre-specified trip thresholds within a period, creating upward notches in their budget sets; Uber randomly assigned drivers to high and low bonus amounts. Here, variation in access to Lyft arises from differences in vehicle-age requirements.

Appendix Figure A9 plots the trip-count densities by assigned threshold (Panel A) and the high-minus-low bonus difference (Panel B). The figure shows significant bunching at the thresholds, and more among drivers assigned the larger bonus. This confirms that drivers respond to the incentives. We recover trip elasticities using a standard bunching estimator, reported in the structural notch convention (see Appendix E.2 for details). Column 1 of Table 6 shows that non-shifters have an implied trip elasticity of 0.60 and shifters 1.10—levels close to our Earnings Accelerator estimates. Just as in the Earnings Accelerator experiments, drivers with access to an additional outside employer are more elastic to Uber. However, the gap of 0.50 is imprecise (bootstrap standard error 0.42; see Appendix E.4). The IDB data nonetheless provide corroborating evidence that differences in outside options lead to differences in labor supply elasticities.

In the Individual Driver Bonus program, variation in access to Lyft comes from vehicle-age requirements (rather than market-level access to Lyft). A concern with this source of variation is that it proxies for driver wealth: shifters mechanically have newer cars, so the gap could reflect wealth rather than Lyft access. Our Houston data reject this as a potential

³¹Using observed utilization rates in our Houston (pre-Lyft) experiment, a driver who cut idle time in half would increase earnings by roughly 35%.

explanation. Table A15 splits the first Houston experiment sample by vehicle age when Lyft was absent: the “pseudo-shifter” and “pseudo-non-shifter” elasticities are economically and statistically indistinguishable ($p > 0.75$ in every specification), consistent with vehicle age not independently predicting labor supply elasticities. Appendix Figure A10 extends this test to vehicle-year quartiles and finds elasticities flat ($p = 0.98$ for joint equality).

5.2 Firm-Specific Elasticities and Markdowns

We can use these estimates to quantify how access to an outside firm changes a firm’s wage-setting power and—under an assumption about market shares—the extent to which workers reallocate hours between firms. The latter is more directly comparable to traditional “quit elasticities” in the literature.

Our estimates indicate that the addition of an outside employer lowers the incumbent firm’s monopsony power. Using the monopsony markdown from equation 1 and the preferred IV-Poisson specification, we find that the implied markdown falls from roughly 68% for drivers without Lyft access to 50%–54% for drivers with it (Table 7).³² In levels, drivers capture only about a third of their marginal revenue product without an outside employer and roughly one-half with one. Even if Uber does not set wages to exploit this power, the firm-specific elasticity remains informative about its market power—which a firm may instead exercise over non-wage amenities, effort, or other margins (Caldwell, Dube, and Naidu, Forthcoming).

The reduction in implied markdowns follows directly from the shifter–non-shifter elasticity gap and requires no assumption about market shares. However, translating the elasticity gap into a cross-platform substitution elasticity requires more structure. This substitution is informative because it tells us how drivers reallocate hours between platforms.

The intensive-margin decomposition (equation 3) writes a shifter’s inside-hours (log-hours)

³²The IV-Tobit estimates imply markdowns of 51% without and 36% with the outside option (Appendix Table A5); Appendix Table A16 reports markdowns across market shares. Because the firm-specific elasticity can differ across wage levels (Caldwell, Dube, and Naidu, Forthcoming), the implied markdown is a local object evaluated at the prevailing wage.

elasticity as $\varepsilon_{\text{shift}} = (1/\phi)\varepsilon - ((1-\phi)/\phi)\sigma$, where ε is the intensive market-hours elasticity, σ is the firm-substitution elasticity, and ϕ is the share of a driver’s ride-share hours spent on Uber. Because every object in this decomposition is an intensive-margin object, we recover $|\sigma|$ from the intensive-margin (log-hours) elasticities, using the non-shifter log-hours estimate for ε : $|\sigma| = (\varepsilon_{\text{shift}} \phi - \varepsilon)/(1 - \phi)$.³³

Because our experiments do not identify ϕ , we report $|\sigma|$ across a range of values, considering four external benchmarks. At the low end, suppose drivers value Uber and Lyft symmetrically and sort across them. Because two-thirds of drivers drive for both platforms (Campbell, 2022), a driver observed on Uber then spends about two-thirds of her ride-share hours there—above one-half because the remaining third drive Uber exclusively—giving $\phi = 0.67$ and $|\sigma|$ of 0.37–0.94. A second benchmark uses Uber’s revealed share of activity: the same source puts Uber at roughly 72% of the US ride-share market (Campbell, 2022), so $\phi = 0.72$ and $|\sigma|$ is 0.58–1.31. A third anchor comes from Aucejo, Perry, and Zafar (2024), who find that 83% of the Uber drivers in their sample drive exclusively for Uber, rather than splitting hours across platforms. At the conservative $\phi = 0.83$, $|\sigma|$ is 1.48–2.86, in line with the cross-firm estimates of Dube, Manning, and Naidu (2025) for the offline economy. At the high end, Koustas (2018) find, using linked financial-account data, that 93% of a typical Uber driver’s ride-share earnings come from Uber; at $\phi = 0.93$, $|\sigma|$ is 4.75–8.52. Across these anchors, cross-platform substitution is economically meaningful: $|\sigma|$ rises from below one at low ϕ , to roughly one and a half to three under our preferred specification ($\phi = 0.83$; preferred estimate 1.48), and to the high single digits as ϕ approaches one.

These calculations assume drivers can freely choose both their total hours and the allocation of their hours across platforms. However, where workers instead hold a single job, the relevant margin is extensive: a worker substitutes by quitting one employer for another. Using the participation elasticities alone (0.19 without Lyft access and 0.37 with it, Column 1 of Table 5), the implied markdowns are 84% and 73%—outside options still reduce the markdown, but by a smaller 11 percentage points. The ability to shift hours freely be-

³³Using the total-response (Poisson) elasticities instead gives $|\sigma|$ of 4.74 (Houston) and 6.64 (pooling Boston) at $\phi = 0.93$ —nearly identical to the intensive recovery for Houston.

tween employers places an additional constraint on employers' wage-setting. Appendix B develops the single-job case in more detail.

Our setting is, in several respects, one where workers should find it easy to respond to pay: Uber and Lyft are co-located competitors offering near-identical work, drivers can multi-app in real time, and pay is salient. Yet low frictions and many employers need not imply high firm-specific elasticities. Dube et al. (2020) document substantial monopsony power on Amazon Mechanical Turk—a platform with many requesters and easily searched tasks—because idiosyncratic tastes over differentiated tasks give each employer market power. In ride-share, the outside option is not a differentiated task but the same job at a near-identical firm. However, adjustment is still costly. This is reflected both in our elasticities and in the fact that a market of paid third-party tools has emerged to help drivers manage earnings across platforms. Read together, our high elasticities and the low elasticity on task platforms are not in tension. Both reflect the same mechanism: firm-specific labor supply rises in the substitutability of workers' outside options—low where the alternative is a differentiated task, high where it is a near-identical firm.

5.3 Sex and Firm-Specific Elasticities

In her 1933 book, Robinson (1933) suggested monopsonistic price discrimination as one explanation for the gender wage gap: if women supply labor less elastically to the firm than men, a wage-setting employer optimally pays them less for equal work. Women might supply labor less elastically to the firm for several reasons: they may be more attached to their employer or face higher switching costs, have less information about their outside options, value employer-provided amenities differently, or face smaller effective labor markets owing to differences in commuting (Black, 1995; Babcock and Laschever, 2009; Manning, 2011). Many of these channels are muted in our setting—men and women drive for the same platform, under the same pay algorithm, with no commute to a fixed workplace—which lets us isolate behavioral factors that could lead to a gap between men and women. However, Section 4 showed that there is such a behavioral gap at the market level that we

cannot attribute to observable differences in characteristics.

Table 8 reports firm-specific elasticities by sex and access to Lyft. For both men and women, access to an outside option raises the elasticity, as in Section 5.1. Among drivers with Lyft access, the firm-specific elasticities are similar—if anything slightly higher for women (1.10 versus 1.00, Poisson), implying a monopsony markdown on women of 48% against 50% for men—and the participation response again rises more for women (a shifter–non-shifter gap on the extensive margin of 0.26 versus 0.07 for men), consistent with the reservation-wage mechanism. We find no sign that women are less elastic to ride-share firms—if anything our point estimates run the other way.

The two components of this elasticity, however, differ by sex. The gap in elasticities for workers with and without access to Lyft is smaller for women than for men (though the difference is not statistically significant). At every benchmark market share, women’s implied cross-platform substitution elasticity is lower than men’s—0.35 versus 1.14 at $\phi = 0.67$, and lower at every other benchmark (lower panel of Table 8). Women reallocate hours between platforms less readily than men. Their total firm-specific elasticity is no lower because a larger participation response offsets this.

Our results are naturally specific to the market we study: ride-share. These drivers face the same pay algorithm and have the same access to Lyft (the same outside options). In other labor markets, sex differences in outside options may arise through homophily in job-search networks or through geographic and commuting constraints that narrow women’s effective labor markets (Caldwell and Danieli, 2024; Black, 1995; Manning, 2011). Our setting simply shows that, in the absence of these constraints, women are no less elastic to their employer overall. This contrasts with what we found at the market level, where women were more elastic.

6 Conclusion

This paper uses randomized wage variation, paired with quasi-experimental variation in access to a competing platform, to estimate labor supply elasticities to the firm and market, separately for men and women. We first show, theoretically, how firm-specific elasticities relate to workers’ outside options, and how monopsonistic markdowns are determined when workers can choose hours flexibly. We then show that, at the market level, a 10% increase in the net-of-commission rate raises ride-share hours by roughly 5–6%; women are about twice as responsive as men and this is not explained by differences in observable characteristics.

When drivers have access to a competing platform, the firm-specific labor supply elasticity rises and the firm-optimal markdown falls. Our preferred estimates indicate that the implied markdown falls from roughly 68% for drivers without Lyft access to 50%–54% for drivers with it. Under our preferred specification, the implied cross-platform substitution elasticity is 1.48–2.86, rising to 4.75–8.52 under higher assumed market shares. More generally, the fact that measured elasticities increase when there is an outside employer highlights the importance of market structure when interpreting data from a single firm or platform: measured responses depend on outside options.

Our results contribute to a small experimental literature that has estimated firm-specific labor supply elasticities; we provide novel estimates, broken down by sex. Our revealed-preference estimates far exceed the near-0.1 elasticity on Amazon Mechanical Turk (Dube et al., 2020) and align with the larger stated-preference experimental estimates of Caldwell, Haegele, and Heining (2025) and Dube, Naidu, and Reich (2022)—a spread consistent with our mechanism, in which firm-specific elasticities rise with the substitutability of workers’ outside options.³⁴ Of course, like most experiments, ours generate temporary variation in pay, so the estimates are short-run; long-run firm-specific elasticities may differ (Bassier and Manning, 2025).

The estimates also speak to sex differences in labor supply—and the sources of the

³⁴Caldwell, Dube, and Naidu (Forthcoming) summarize this literature.

gender wage gap. We find significant sex differences at the market level: women are about twice as elastic as men to the ride-share market. Observable characteristics and driving patterns do not explain them. Yet, women are no less elastic to their employer. We find no support for the classic monopsonistic-discrimination account of the gender wage gap: the implied markdown on women (48%) is no larger than on men (50%), so this channel explains essentially none of the gap. This null is not mechanical: the gender gap in elasticity to the market is real and unexplained by observables. In other labor markets, commuting constraints, search networks, or information may leave women less elastic to their employer; our setting holds these channels fixed, isolating behavior.

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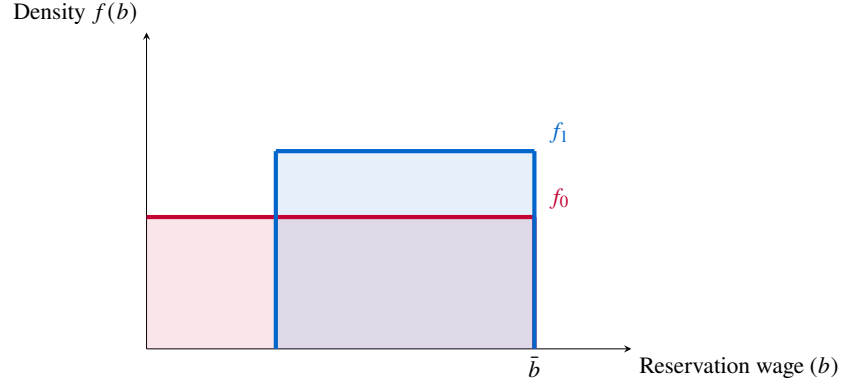
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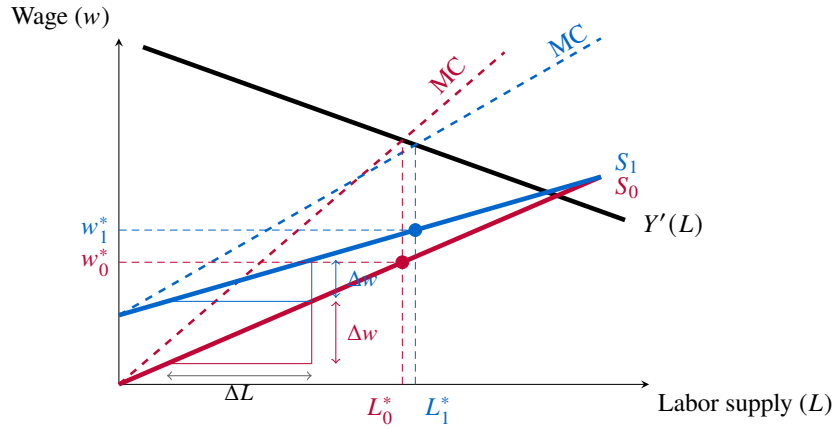
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7 Tables and Figures

Figure 1: Outside Options and Labor Supply
Panel A: Distribution of Outside Options

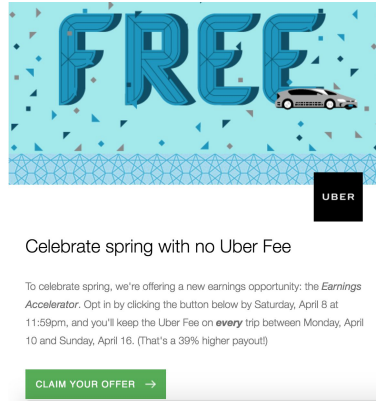


Panel B: Labor Supply to the Firm



Note: Worker i accepts wage w when her reservation wage satisfies $b_i \leq w$, so $L(w) = N \cdot F(w)$ and the supply curve is $w(L) = F^{-1}(L/N)$. Panel A: f_0 (red) is uniform on $[0, \bar{b}]$; f_1 (blue) is uniform on a higher support and first-order stochastically dominates f_0 . Panel B: the FOSD shift flattens the supply curve, raising the equilibrium wage from w_0^* to w_1^* .

Figure 2: Earnings Accelerator Messaging



Note: This is an example of the message that was sent to drivers selected for the Earnings Accelerator in the main Houston experiment. The link directed the driver to a more detailed opt-in page with information on how the incentive worked and with consent language.

Table 1: Elasticities and Identifying Variation

Parameter	Object	Identified from	Source of variation
$\varepsilon, \bar{\varepsilon}$	Market hours (intensive, total) ($\tau_{\text{non}} = \bar{\varepsilon}$)	Non-shifters	Earnings Accelerator, Houston Spring 2017 (Lyft absent)
τ_{shift}	Firm-specific (total)	Shifters	Earnings Accelerator, Houston Fall 2017 / Boston (Lyft present)
$ \sigma $	Cross-firm	Log-hours gap ($\varepsilon_{\text{shift}}$ vs. ε)	Shifter vs. non-shifter comparison, given ϕ

Note: This table links each elasticity in the conceptual framework (Section 2) to its source of identifying variation. ε is the intensive market-hours elasticity and $\bar{\varepsilon} = \eta^{\text{firm}} + \varepsilon$ the participation-inclusive total, both estimated from drivers without access to a competing platform; for these drivers the firm-specific total coincides with $\bar{\varepsilon}$ ($\tau_{\text{non}} = \bar{\varepsilon}$). τ_{shift} is the total firm-specific elasticity, estimated from drivers with access to Lyft. The reported $|\sigma|$ is the magnitude of the elasticity of outside hours to the inside wage, recovered from the gap between the shifter and non-shifter log-hours (intensive) elasticities using Equation 3 and a range of assumed inside market shares ϕ (reported in Section 5.2).

Table 2: Descriptive Statistics

	Houston 1 (1)	Houston 2 (2)	Boston (3)	IDB (4)
Female	0.52	0.39	0.14	0.18
Age	44.1 (12.2)	—	—	40.3 (11.7)
Vehicle year	2013.5 (2.5)	2013.6 (2.7)	2010.5 (4.5)	2013.6 (3.4)
Months on platform	16.0 (10.2)	8.7 (9.4)	11.1 (8.7)	14.6 (13.1)
Hours per week (pre-period)	19.6 (13.6)	21.5 (12.9)	—	—
Vehicle Solutions	0.12	0.08	0.08	0.00
Drivers	2,020	2,100	1,600	58,928

Note: This table compares driver characteristics across the four experimental samples. Houston 1 (Spring 2017) is a non-shifter sample because Lyft had temporarily withdrawn from the market. Houston 2 (Fall 2017) includes shifters, as Lyft had returned. Boston (Aug–Oct 2016) is a mixed sample in which 88% of drivers were eligible for Lyft. The IDB sample (San Francisco, 2017) covers drivers in the Individual Driver Bonus program. “Shifter” indicates whether a driver’s vehicle qualifies for Lyft. Age is unavailable for Houston 2 and Boston. Appendix Table A17 presents Houston 1 characteristics by sex.

Table 3: Market-Level Elasticities

	1{Drive} (1)	log(Hours) (2)	(3)	Poisson (4)	(5)
Elasticity	0.19*** (0.05)	0.42*** (0.09)	0.69*** (0.09)	0.47*** (0.09)	0.64*** (0.07)
Observations	3,108	2,766	4,650	3,108	5,154
Fee-Free Driving Data	✓	✓	✓	✓	✓
Taxi Data			✓		✓
Strata Dummies	✓	✓	✓	✓	✓
Baseline Covariates	✓	✓	✓	✓	✓

Note: This table presents pooled estimates of Equation 5. The dependent variable in Column 1 is an indicator for non-zero hours; Columns 2–3 use log hours worked; Columns 4–5 (Poisson) use the level of hours. The endogenous variable throughout is $\log(1 - \kappa_{it})$. We scale the coefficient in Column 1 by the baseline participation rate so that we recover an elasticity rather than a semi-elasticity. Columns 1–3 estimate the model via two-stage least squares; Columns 4–5 use a Poisson model. An IV-Tobit specification is reported in Appendix Table A5. The baseline covariates include months on the platform, use of the “vehicle solutions” program, and one lag of log earnings. Table 4 reports market-level estimates separately by sex. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table 4: Market-Level Labor Supply Elasticities by Sex

	1{Drive}	log(Hours)		Poisson	
	(1)	(2)	(3)	(4)	(5)
Male	0.11 (0.09)	0.28* (0.15)	0.43*** (0.14)	0.29** (0.14)	0.42*** (0.12)
Female	0.25*** (0.07)	0.53*** (0.11)	0.88*** (0.11)	0.62*** (0.11)	0.81*** (0.10)
<i>p</i> -value (equal)	[0.217]	[0.171]	[0.013]	[0.061]	[0.012]
First Stage <i>F</i> : Male	759.2	745.8	897.5	—	—
First Stage <i>F</i> : Female	1583.7	1694.4	1669.1	—	—
Observations	3,108	2,766	4,650	3,108	5,154
Fee-Free Driving Data	✓	✓	✓	✓	✓
Taxi Data			✓		✓
Strata Dummies	✓	✓	✓	✓	✓
Baseline Covariates	✓	✓	✓	✓	✓

Note: Market-hours labor-supply elasticities by sex, on the extensive margin (1{Drive}), scaled by the baseline participation rate), the intensive margin (log hours, 2SLS), and total (Poisson), estimated on the Houston non-shifter (pre-Lyft) experiment—fee-free weeks, with the taxi weeks added in the total-response columns—the by-sex analogue of Table 3. The endogenous variable is $\log(1 - \kappa_{it})$; all models control for randomization strata and baseline covariates interacted with sex. The *p*-value (equal) row reports the test that the male and female elasticities coincide. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table 5: Labor Supply with Multiple Firms

	1{Drive}		Log(Hours)		Poisson Model	
	(1)	(2)	(3)	(4)	(5)	(6)
Non-Shifter	0.19*** (0.06)	0.19*** (0.05)	0.42*** (0.09)	0.42*** (0.09)	0.47*** (0.09)	0.48*** (0.09)
Shifter	0.37*** (0.13)	0.26*** (0.06)	0.81*** (0.21)	1.10*** (0.11)	0.87*** (0.18)	1.01*** (0.12)
<i>p</i> -value (equal)	[0.202]	[0.408]	[0.091]	[0.000]	[0.051]	[0.000]
Observations	4,700	7,172	4,155	6,369	4,700	7,172
Houston EA	✓	✓	✓	✓	✓	✓
Boston EA		✓		✓		✓

Note: The endogenous variable throughout is $\log(1 - \kappa_{it})$. The dependent variable in Columns 1–2 is an indicator for non-zero hours; the dependent variable in Columns 3–4 is the log of hours worked. Columns 5–6 present a Poisson regression model. In Columns 1–2 we scale the coefficients by baseline participation rates so that they can be interpreted as elasticities. All models control for the strata used for random assignment and for baseline covariates, interacted with whether a driver is a “shifter”. The model uses dummies for each treatment offer. Standard errors are clustered by driver. See Section 5.1 for details and Table A13 for additional robustness checks. Levels of significance: *10%, ** 5%, and *** 1%.

Table 6: Labor Supply with Multiple Firms: Evidence from the IDB Program

	(1)	(2)	(3)	(4)	(5)
Non-Shifter	0.602*	0.419	0.859**	0.895***	0.916**
	(0.364)	(0.283)	(0.400)	(0.182)	(0.359)
Shifter	1.100***	1.062***	1.399***	1.349***	1.314***
	(0.188)	(0.177)	(0.210)	(0.174)	(0.178)
Shifter – Non-Shifter	0.498	0.643**	0.539	0.454*	0.398
	(0.416)	(0.327)	(0.443)	(0.267)	(0.399)
Elasticities	216	216	227	221	220
Driver-Periods	878,323	878,323	965,714	943,769	934,926
Threshold Dummies	✓	✓	✓	✓	✓
Sample-Size Weighted	✓		✓	✓	✓
Width	50	50	50	100	100
Degree	3	3	3	3	4
Excluded Bins	4	4	3	3	3

Note: This table presents estimates of drivers' trip elasticities using data from the Individual Driver Bonus Program. Each column reports a regression of cell-level bunching elasticities on a shifter indicator, weighted by the number of drivers in each cell except in Column 2, which is unweighted. "Non-Shifter" is the constant (non-shifter mean); "Shifter" is the non-shifter mean plus the shifter increment. Column 1 is our preferred specification; Column 2 is unweighted; Columns 3–5 vary the bunching estimation parameters. Estimates with $|\varepsilon| > 10$ are trimmed. All regressions include threshold fixed effects. Robust standard errors in parentheses. See Appendix E.2 for details on the bunching estimation procedure and Appendix E.4 for standard error computation. Levels of significance: *10%, ** 5%, and *** 1%.

Table 7: Firm-Specific Elasticities

	Earnings Accelerator	
	Houston (1)	+ Boston (2)
Non-Shifter (τ_{non})	0.47*** (0.09)	0.48*** (0.09)
Shifter (τ)	0.87*** (0.18)	1.01*** (0.12)
Markdown $1/(1 + \tau)$, Non-Shifter	67.8% (4.0%)	67.8% (4.0%)
Shifter	53.5% (5.2%)	49.7% (2.9%)
<i>Implied σ by market share ϕ, from intensive-margin elasticities (SE in parentheses):</i>		
$\phi = 0.67$ (symmetric tastes)	0.37 (0.51)	0.94 (0.35)
$\phi = 0.72$ (Uber market share)	0.58 (0.63)	1.31 (0.43)
$\phi = 0.83$ (Aucejo et al.)	1.48 (1.16)	2.86 (0.76)
$\phi = 0.93$ (Koustas)	4.75 (3.10)	8.52 (1.95)
Houston EA	✓	✓
Boston EA		✓

Note: This table uses the structural model in Section 2 to recover the implied firm-substitution elasticity $|\sigma|$ from the shifter and non-shifter estimates. Non-shifter estimates identify $\tau_{\text{non}} = \bar{\varepsilon}$, the participation-inclusive total market-hours elasticity; shifter estimates identify the firm-specific total elasticity τ_{shift} , which also reflects substitution from Lyft. The elasticity and markdown columns use the IV-Poisson (total) estimates from Table 5; the implied $|\sigma|$ rows use the intensive-margin (log-hours) estimates from the same table. Markdown is $1/(1 + \tau)$ and does not depend on ϕ . We report $|\sigma|$ at four benchmark values of the inside market share ϕ : symmetric driver tastes conditional on driving for Uber ($\phi = 0.67$), Uber's share of the US ride-share market ($\phi = 0.72$; Campbell, 2022), the lower bound implied by the single-homing rate in Aucejo, Perry, and Zafar (2024) ($\phi = 0.83$), and the external calibration of Koustas (2018) ($\phi = 0.93$). The framework requires $\phi \geq \varepsilon/\varepsilon_{\text{shift}}$, below which the implied $|\sigma|$ is non-positive. Standard errors on the elasticities, markdowns, and $|\sigma|$ are computed via the delta method, treating the shifter and non-shifter estimates as independent (they come from non-overlapping samples). Levels of significance: *10%, ** 5%, and *** 1%.

Table 8: Firm-Specific Labor Supply Elasticities by Sex

	Male		Female	
	1{Drive} (1)	Poisson (2)	1{Drive} (3)	Poisson (4)
Non-Shifter (τ_{non})	0.11 (0.09)	0.30** (0.14)	0.26*** (0.07)	0.62*** (0.11)
Shifter (τ_{shift})	0.18*** (0.07)	1.00** (0.44)	0.52*** (0.15)	1.10*** (0.27)
<i>Implied σ by market share ϕ, from total-response (Poisson) elasticities:</i>				
$\phi = 0.67$ (symmetric tastes)		1.14		0.35
$\phi = 0.72$ (Uber market share)		1.52		0.61
$\phi = 0.83$ (Aucejo et al.)		3.16		1.72
$\phi = 0.93$ (Koustas)		9.10		5.74

Note: Firm-specific labor-supply elasticities by sex and access to Lyft, on the extensive margin (1{Drive}, scaled by the baseline participation rate) and total (Poisson). Non-shifters lack access to Lyft (Houston 1) and shifters have it (Houston 2 and Lyft-eligible Boston drivers)—the by-sex analogue of Table 5. The endogenous variable is $\log(1 - \kappa_{it})$; all models control for randomization strata and baseline covariates interacted with sex. The lower panel translates the shifter and non-shifter estimates into the implied firm-substitution elasticity $|\sigma|$ at the four market-share benchmarks of Table 7, separately by sex, using the total-response (Poisson) elasticities in the upper panel; this makes the male and female substitution elasticities directly comparable and verifiable from the estimates above. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Online Appendix for “Outside Options and Labor Supply: Evidence from the Gig Economy”

Sydnee Caldwell and Emily Oehlsen

A Appendix Tables and Figures

Figure A1: Pay on Uber

Weekly Earnings	
▼ Trip Earnings	\$19.34
Fare	\$24.12
Uber Fee	- \$6.03
Toll	+ \$1.25
Estimated Payout	\$19.34

Note: This figure shows that, when we ran the experiment, drivers’ weekly pay statements listed (1) how much they collected in trip receipts, (2) how much of this went to Uber in the form of the Uber fee, (3) what, if anything, they earned in Uber promotions, and (4) what, if any, reimbursements they received for tolls. Their estimated payout was the sum of these four items. The structure of drivers’ weekly earnings statements has since changed.

Figure A2: Example of a Third-Party Multi-Apping Application



Note: This is a screenshot of a TechCrunch article discussing a third-party app, Mystro, which helps drivers quickly switch between competing ride-share platforms. This is one of many third-party apps that help drivers maximize earnings through multi-apping. As of August 10, 2019 (before the COVID-19 pandemic), Mystro charged \$5/week, \$12/month, or \$100/year for this service.

A3

Figure A3: Multi-Apping Information from Driver Forums and Guides

Panel A: Driver Forum Discussion

Panel B: Multi-Apping Guide



How to run the apps at the same time

Once you have everything set up you will need to actually open and run both apps at the same time. This is pretty easy for the most part, but there are a few tricks that will help you out.

First, close out any apps you have open. Driving for Lyft and Uber at the same time requires you to keep both apps open until you get a request, which puts quite a bit of strain on your phone, sucks the battery dry, and uses a ton of data in the process. So bring a charger, stay on task, and be prepared to pull down a lot of data. And I mean, a TON.

Second, try to minimize surfing the web or being active on social media. Keep in mind you have two apps that are constantly talking to their respective platforms, so the chance of your phone crashing or glitching out becomes much higher than usual. The last thing you want is to get a ride request and have your phone freeze up or crash, costing you both time and money.

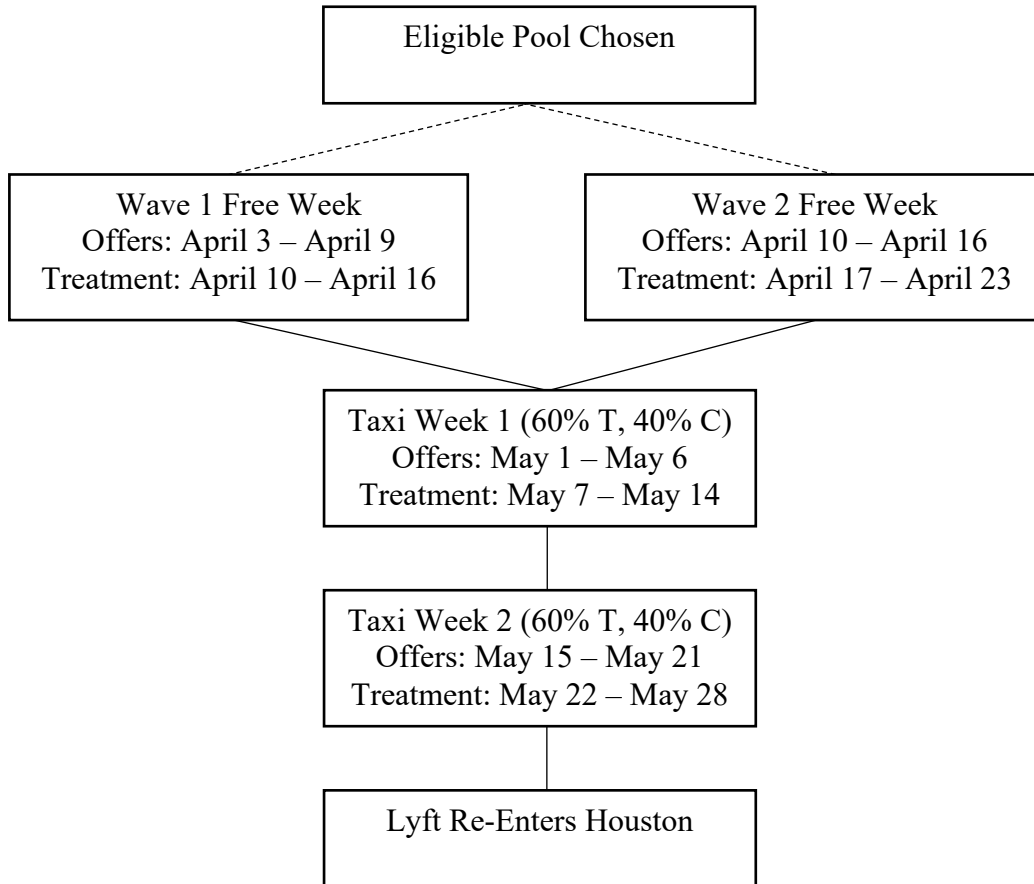
It is also worth noting that when you have both of the apps open, you will need to have Uber open on the main screen, with Lyft running in the background behind it. Uber will automatically close out after a minute or two if it is running in the background, but Lyft will stay open.

Accepting ride requests

When you get a ride request, make sure to accept one and immediately log right out of the other. It won't take long to get a ride request, and on a busy night you may get two at the same time. If this happens, pick the one closest to you and decline the other. As every driver knows, the best way to make money is in volume, and the less dead miles you have, the better.

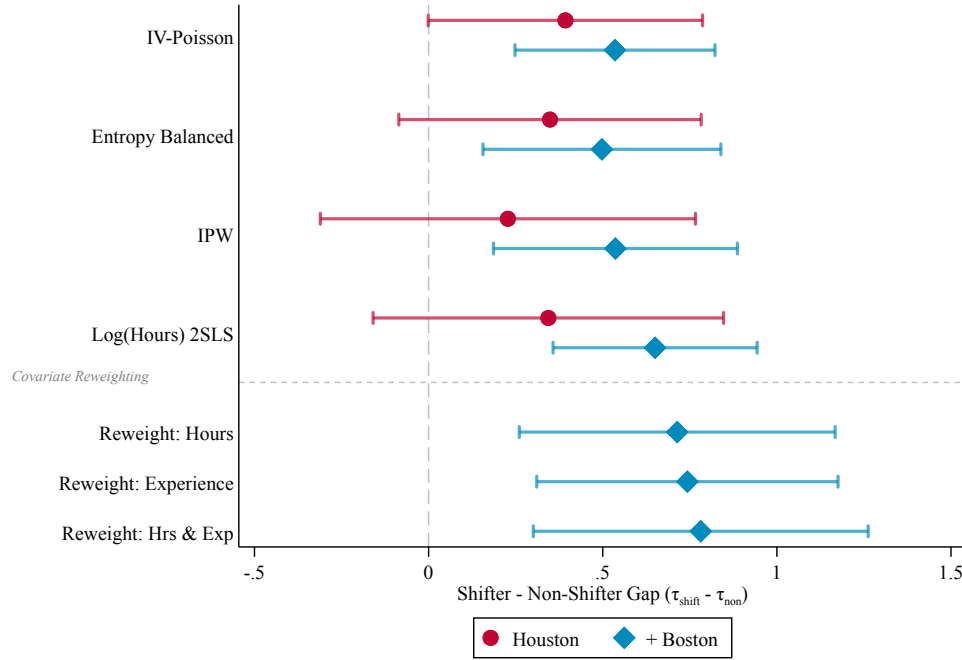
Note: Panel A shows a screenshot from “Uber People,” an online forum where drivers discuss ride-share topics. The forum is not affiliated with Uber Technologies, Inc. or any other ride-share company. The conversation highlights that drivers are interested in multi-apping but find it requires non-trivial effort. Panel B shows a screenshot from rideshareapps.com’s guide on driving for Lyft and Uber simultaneously, illustrating that multiple non-affiliated forums provide information on maximizing earnings via multi-apping.

Figure A4: Earnings Accelerator Timeline Example



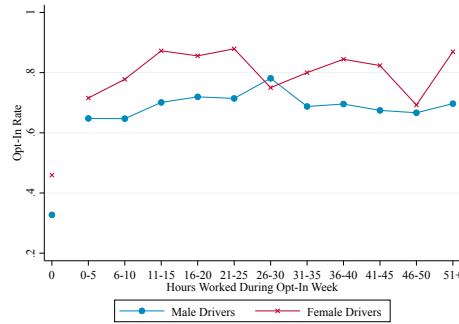
Note: This diagram shows the timeline of the first Houston Earnings Accelerator experiment. All dates are from 2017. Appendix C.2 provides additional detail on the timeline and structure of the second Houston experiment and the Boston experiment.

Figure A5: Robustness of the Shifter–Non-Shifter Gap



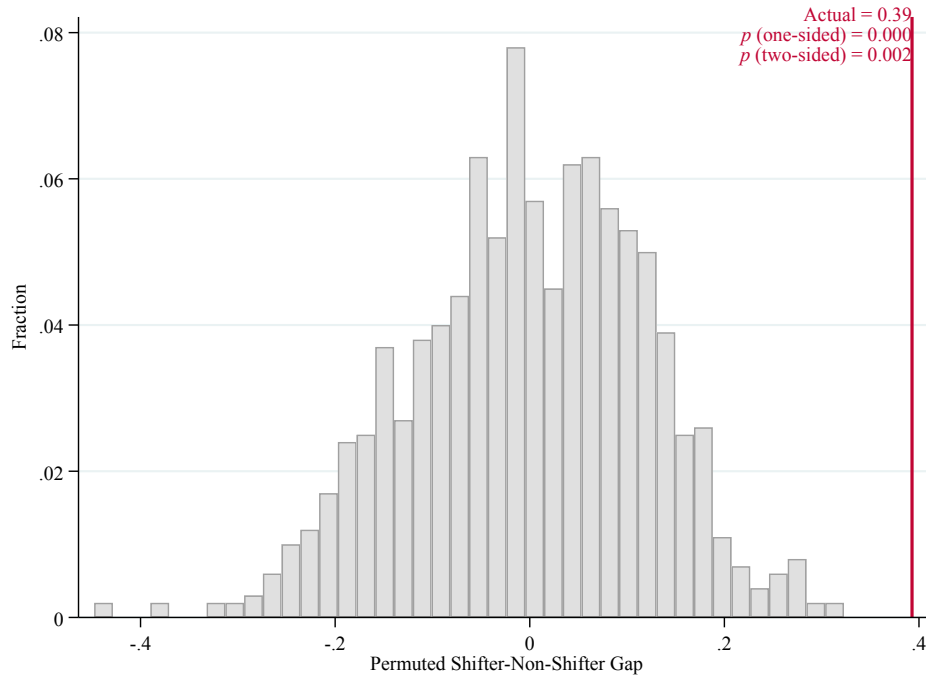
Note: Each point reports the estimated gap $\tau_{\text{shift}} - \tau_{\text{non}}$ between shifter and non-shifter elasticities. Circles show Houston-only estimates; diamonds show pooled Houston and Boston estimates. The top section varies the estimation method: the IV-Poisson model is our preferred specification; entropy balancing reweights non-shifters to match the shifter covariate distribution with exact mean balance; IPW uses inverse probability weights applied to 2SLS; the log-hours 2SLS specification drops non-workers. The bottom section allows elasticities to vary by covariate bin and reweights using the non-shifter covariate distribution, showing that the gap is not driven by compositional differences in hours worked or experience between shifters and non-shifters. Whiskers denote 95% confidence intervals.

Figure A6: Opt-In Rates by Opt-In Week Hours



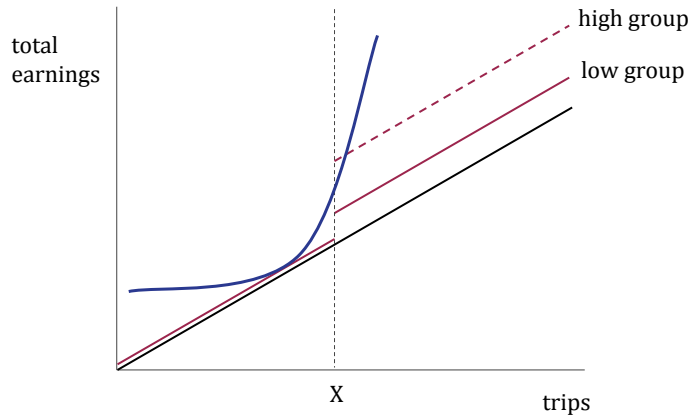
Note: This figure plots the proportion of drivers who accepted our initial (non-taxi) offer during the first Houston experiment, separately by sex and by hours worked during opt-in week. The far left dots plot the proportions for drivers who drove 0 hours during opt-in week. The remaining dots plot proportions for drivers whose hours fell within the intervals $(a, b]$ as indicated.

Figure A7: Permutation Inference for the Shifter–Non-Shifter Gap



Note: This figure plots the distribution of the shifter–non-shifter elasticity gap from 1,000 random permutations of shifter status across the pooled Houston sample. Each permutation randomly re-assigns the shifter label (preserving the total number of shifters), reconstructs the interaction terms and instruments, and re-estimates the preferred IV-Poisson specification (Column 5 of Table 5). The vertical line marks the actual gap (0.40). The one-sided p -value is below 0.001; the two-sided p -value is 0.002.

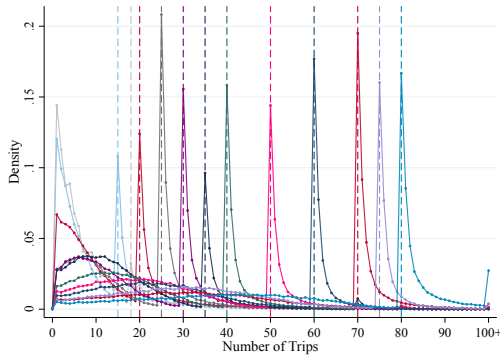
Figure A8: Individual Driver Bonus Budget Sets



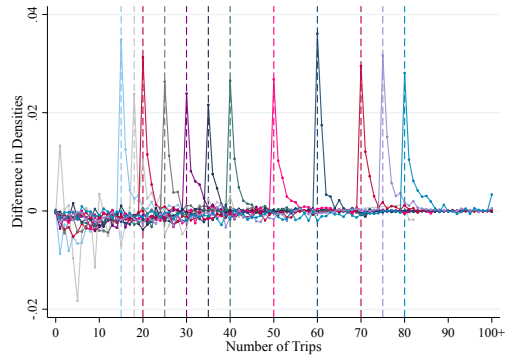
Note: This figure shows a stylized budget set for the Individual Driver Bonus (IDB) incentive. For a given trip threshold X , drivers in the IDB received either a high or a low lump-sum bonus for exceeding the trip threshold. The blue curve shows an indifference curve for a driver who would exceed the threshold if offered the high, but not the low, bonus.

Figure A9: Bunching in the Individual Driver Bonus Program

A. Bunching at Trip Thresholds

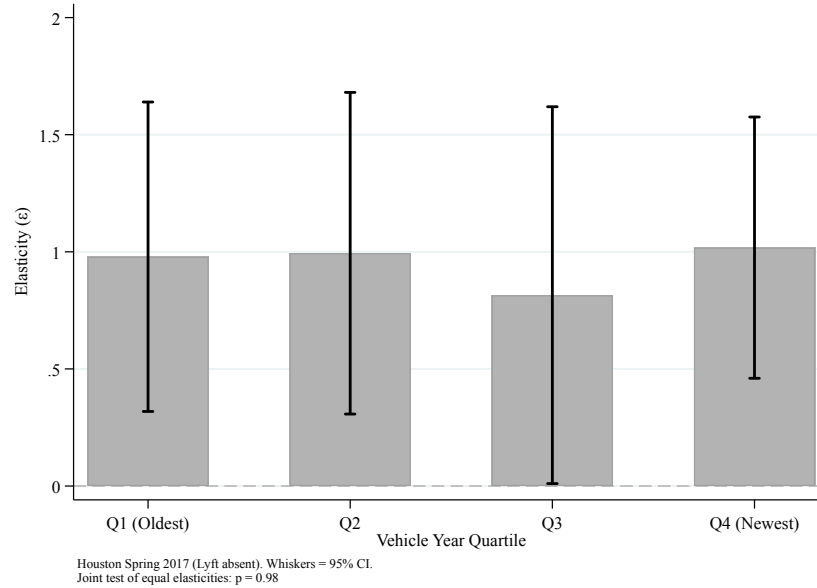


B. High-Low Difference in Bunching



Note: This figure shows that the IDB incentive leads to bunching at the assigned trip thresholds. Panel A shows the raw density of trips completed by assigned trip threshold, marked by a dashed line of the same color. Panel B shows the difference in densities between the drivers randomly assigned the high bonus and the drivers randomly assigned the low bonus for a given trip threshold. Densities are calculated separately by assigned trip threshold (shown as dashed lines in the same color).

Figure A10: Vehicle Year Gradient in Houston 1 (Placebo)



Note: This figure estimates labor supply elasticities by vehicle year quartile in the Houston Spring 2017 experiment, when Lyft was absent from the market. If vehicle age independently predicted differential elasticities, we would expect a gradient across quartiles. The estimates are flat ($p = 0.98$ for the joint test of equality), consistent, over the range of vehicle ages we observe, with the shifter–non-shifter gap in the Houston Fall 2017 and Boston experiments reflecting access to Lyft rather than vehicle characteristics. Whiskers show 95% confidence intervals.

Table A1: Data Sources

	(1) Boston EA	(2) Houston EA 1	(3) Houston EA 2	(4) San Francisco IDB
City	Boston	Houston	Houston	San Francisco
Timeline	Aug–Oct 2016	Apr–May 2017	Sep–Oct 2017	Jul–Dec 2017
Sample Size	1,600	2,020	2,100	58,928
Shifter Definition	Car Year ≥ 2004	No Drivers	All Drivers	Car Year ≥ 2004
Market-Hours Elasticity	×	✓	×	×
Firm Substitution	✓	✓	✓	✓

Note: This table describes the four data sources used in the paper. Columns 1–3 describe the three Earnings Accelerator (EA) experiments, which offered drivers a week of fee-free driving. Column 4 describes the Individual Driver Bonus (IDB) program, which offered drivers lump-sum bonuses for exceeding trip thresholds. “Shifter Definition” describes how drivers are classified as having access to an outside employment option in each sample. In the Boston EA and the IDB, shifters are drivers whose vehicle is new enough to qualify for Lyft. The Houston 1 experiment includes only non-shifters and Houston 2 includes only shifters. More information on the EA experiments is provided in Appendix Section C.2; details on the IDB sample are in Appendix Section E.1.

Table A2: Earnings Accelerator Offers and Opt-In Decisions

	Offered $\log(1 - \kappa)$		$1\{\text{Accepted}\}$		$\log(1 - \kappa)$		$\log(W)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Pooled</i>								
Treated	0.29*** (0.001)	0.29*** (0.001)	0.67*** (0.010)	0.59*** (0.008)	0.20*** (0.003)	0.17*** (0.002)	0.16*** (0.009)	0.17*** (0.008)
Observations	4,040	6,746	4,040	6,746	4,040	6,746	2,998	5,031
<i>Panel B. By Sex</i>								
Male	0.28*** (0.002)	0.28*** (0.002)	0.61*** (0.016)	0.57*** (0.012)	0.17*** (0.005)	0.16*** (0.004)	0.12*** (0.012)	0.14*** (0.011)
Female	0.31*** (0.001)	0.31*** (0.001)	0.73*** (0.014)	0.60*** (0.011)	0.22*** (0.004)	0.19*** (0.003)	0.19*** (0.012)	0.20*** (0.012)
Observations	4,040	6,746	4,040	6,746	4,040	6,746	2,998	5,031
Strata Dummies	✓	✓	✓	✓	✓	✓	✓	✓
Fee-Free Driving Data	✓	✓	✓	✓	✓	✓	✓	✓
Taxi Data		✓		✓		✓		✓

Note: This table presents estimates of the impact of the treatment on the offered log net-of-commission rate (Columns 1–2), opt-in decisions (Columns 3–4), the realized log net-of-commission rate (Columns 5–6) and realized log hourly earnings (Columns 7–8). We use a stacked model, controlling for the strata used for random assignment. Panel A pools all drivers and includes a female dummy; Panel B reports estimates separately by sex. Odd columns use data from the two weeks of “fee-free” driving; even columns add data from the two weeks of taxi experiments. Standard errors, clustered by driver, are presented in parentheses. Levels of significance: *10%, ** 5%, and *** 1%.

Table A5: Elasticities under the IV-Tobit Specification

	Houston	Houston + Boston
	(1)	(2)
Non-Shifter (τ_{non})	0.96*** (0.17)	0.96*** (0.17)
Shifter (τ_{shift})	1.78*** (0.43)	1.76*** (0.21)
p -value (equal)	[0.074]	[0.003]
Implied markdown, non-shifter	51%	51%
Implied markdown, shifter	36%	36%
Observations	4,700	7,172

Note: IV-Tobit estimates of the firm-specific labor supply elasticity for non-shifters and shifters—the analogue of the Poisson estimates in Table 5. The IV-Tobit accounts for censoring at zero hours under latent normality and a chosen censoring threshold; we report it as a robustness check and do not rely on it. The pooled overall market-hours elasticity under IV-Tobit is 0.96 (fee-free weeks) to 1.05 (including taxi-driving data). Markdowns are $1/(1 + \tau)$. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A3: Earnings Accelerator Balance

	Boston		Houston Experiments			
	Sample Mean	Week 1-2 Balance	Experiment 1		Experiment 2	
			Wave 1- Wave 2	Taxi Treated-Control	Sample Mean	Treatment-Control
	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.14	-0.02 (0.019)	0.02 (0.020)	0.01 (0.017)	0.39	0.01 (0.020)
Months on Platform	11.14	-0.01 (0.250)	0.30 (0.212)	0.06 (0.184)	8.75	-0.19 (0.250)
Vehicle Solutions	0.08	-0.00 (0.016)	0.01 (0.014)	-0.01 (0.013)	0.08	0.01 (0.012)
Vehicle Year	2009.27	0.13 (0.192)	0.03 (0.111)	-0.10 (0.096)	2013.63	0.07 (0.119)
Prior Week Hours	13.87	-0.01 (0.690)	-0.37 (0.565)	-0.52 (0.619)	14.82	0.71 (0.619)
Prior Week Earnings	242.90	-2.92 (12.325)	-6.88 (8.973)	-9.69 (8.942)	205.76	3.75 (8.444)
F-statistic		0.48	0.70	0.48		0.82
p-value from F-test		0.82	0.65	0.83		0.55
Observations		1,600	2,020	2,710		2,100

Note: This table presents randomization balance for the three Earnings Accelerator experiments. Columns 1–2 present the sample mean and strata-adjusted week 1 vs. week 2 balance for the Boston experiment. Columns 3–4 show strata-adjusted differences for the first Houston (non-shifter) experiment: between drivers in waves 1 and 2 of the fee-free phase (Column 3) and between drivers offered and not offered a taxi contract (Column 4). Column 4 includes two observations for each of the 1,355 drivers who accepted the initial fee-free offer and were eligible for the taxi phase. Columns 5–6 present the sample mean and strata-adjusted treatment-control balance for the second Houston experiment. Missing vehicle years are coded as zero, which lowers the reported vehicle-year means relative to the driver-level descriptive statistics in Table 2. Levels of significance: *10%, ** 5%, and *** 1%.

Table A4: Non-Experimental Estimates of Labor Supply Elasticities

	Non-Experimental Weeks			Experimental Weeks		
	(1)	(2)	(3)	(4)	(5)	(6)
A. OLS						
Log(W)	0.13** (0.06)	0.11* (0.06)	0.16*** (0.04)	0.20*** (0.07)	0.20*** (0.07)	0.23*** (0.05)
Observations	9,870	9,870	7,651	5,031	5,031	4,648
B. 2SLS (Leave-One-Out Log W)						
Log(W)	-0.07 (0.08)	-0.07 (0.08)	0.08 (0.07)	1.90*** (0.39)	1.90*** (0.39)	2.28*** (0.41)
Observations	9,870	9,870	7,651	5,031	5,031	4,648
Months on Uber		✓	✓		✓	✓
Lag of Log(Hours)			✓			✓

Note: This table reports estimates from regressions of log hours on log wages (earnings/hours worked), pooling male and female drivers. Panel A reports OLS estimates. Panel B instruments log wages with the leave-one-out mean of wages among all other drivers in the same week. Because the instrument varies at the week level, the IV specifications exclude week fixed effects; the OLS specifications include them. Standard errors (in parentheses) are clustered at the driver level. Columns 1–3 use data from non-experimental weeks surrounding our main (pre-Lyft) Houston experiment. Columns 4–6 use data from experimental weeks. Levels of significance: *10%, **5%, and *** 1%.

Table A6: Robustness of Labor Supply Elasticities

	1{Drive}	log(Hours)				Poisson Model	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Just-Identified Model							
Male	0.13 (0.09)	0.47* (0.24)	0.30* (0.16)	0.61*** (0.19)	0.47*** (0.15)	0.39* (0.21)	0.55*** (0.18)
Female	0.29*** (0.07)	0.62*** (0.16)	0.46*** (0.12)	0.95*** (0.14)	0.86*** (0.13)	0.66*** (0.15)	0.90*** (0.15)
<i>p</i> -value (equal)	[0.179]	[0.600]	[0.417]	[0.148]	[0.048]	[0.307]	[0.140]
Observations	4,040	2,998	2,998	5,031	5,031	4,040	6,746
Panel B. Strata Only							
Male	0.12 (0.09)	0.40* (0.23)	0.28* (0.16)	0.55*** (0.19)	0.41*** (0.15)	0.37* (0.21)	0.39** (0.19)
Female	0.31*** (0.07)	0.57*** (0.16)	0.45*** (0.12)	0.93*** (0.14)	0.80*** (0.12)	0.65*** (0.15)	0.75*** (0.15)
<i>p</i> -value (equal)	[0.115]	[0.544]	[0.413]	[0.091]	[0.043]	[0.285]	[0.118]
Observations	4,040	2,998	2,998	5,031	5,031	4,040	6,750
Endogenous Variable	$\log(1 - \kappa)$	$\log(W)$	$\log(1 - \kappa)$	$\log(W)$	$\log(1 - \kappa)$	$\log(1 - \kappa)$	$\log(1 - \kappa)$
Fee-Free Driving Data	✓	✓	✓	✓	✓	✓	✓
Taxi Data				✓	✓		✓
Strata Dummies	✓	✓	✓	✓	✓	✓	✓

Note: This table presents results analogous to those in Table 4, estimated separately by sex. Both panels control only for the strata used for random assignment, dropping the baseline covariates used in Table 3; Panel A uses just-identified instruments and Panel B the over-identified instrument set. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A7: Earnings Elasticities

	1{Earn}	Log(Fares)		Poisson Model	
	(1)	(2)	(3)	(4)	(5)
Panel A. Just-Identified Model					
Male	0.11 (0.09)	0.25 (0.16)	0.45*** (0.15)	0.26* (0.15)	0.43*** (0.12)
Female	0.24*** (0.07)	0.61*** (0.13)	1.01*** (0.12)	0.68*** (0.11)	0.91*** (0.10)
<i>p</i> -value (equal)	[0.290]	[0.080]	[0.005]	[0.022]	[0.002]
Observations	3,108	2,715	4,581	3,108	5,154
Panel B. Over-Identified Model					
Male	0.14 (0.09)	0.25 (0.17)	0.43*** (0.16)	0.32 (0.22)	0.36* (0.20)
Female	0.31*** (0.07)	0.52*** (0.13)	0.90*** (0.13)	0.71*** (0.16)	0.84*** (0.15)
<i>p</i> -value (equal)	[0.164]	[0.218]	[0.025]	[0.159]	[0.047]
First Stage <i>F</i> : Male	212	213	80	—	—
First Stage <i>F</i> : Female	417	533	177	—	—
Observations	4,040	2,944	4,954	4,040	6,750
Fee-Free Driving Data	✓	✓	✓	✓	✓
Taxi Data			✓		✓
Strata Dummies	✓	✓	✓	✓	✓
Baseline Covariates (Panel A)	✓	✓	✓	✓	✓

Note: This table presents results estimated separately by sex, analogous to those in Table 4, where the outcome variable is either an indicator for positive earnings that week (Column 1) or the log of fares collected (Columns 2–3). Columns 4 and 5 present results from Poisson models. Panel A presents results from a just-identified model that uses the same baseline covariates as Table 3. Panel B presents results from the over-identified model, controlling only for the strata used for random assignment. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A8: Treatment Effects During Non-Treatment Weeks

	Before Wave 1		Treatment Weeks		After Wave 2	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. 1{Drive}						
Treated	-0.014 (0.018)	-0.015 (0.018)	0.033*** (0.008)	0.033*** (0.008)	0.012 (0.022)	0.012 (0.022)
Observations	2,020	2,020	4,040	4,040	1,355	1,355
Panel B. log(Hours)						
Treated	-0.040 (0.037)	-0.037 (0.037)	0.087*** (0.022)	0.089*** (0.022)	0.032 (0.044)	0.032 (0.044)
Observations	1,554	1,554	2,998	2,998	1,066	1,066
Strata Dummies	✓	✓	✓	✓	✓	✓
Baseline Covariates		✓		✓		✓

Note: This table examines the impact of the treatment on driving behavior before and after treatment. Each estimate comes from a regression of the indicated variable on a dummy for whether the driver was treated, controlling for the strata used for random assignment (all columns) and for baseline covariates (even columns; sex, vehicle, and tenure—lagged earnings are unavailable outside the treatment weeks and are excluded). The data come from the first Houston experiment, conducted before Lyft re-entered the market. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A9: Heterogeneity in Market-Level Elasticities

	By Months on Platform		By Age		Time of Day			By Hours Strata			
	Top 1/2 (1)	Bottom 1/2 (2)	≤35 (3)	>35 (4)	Late Night (5)	Weekend (6)	School Pick-Up (7)	Low (8)	High (9)	Very High (10)	“Full Time” (11)
Panel A. 1{Drive} (2SLS Model)											
Male	0.09 (0.10)	0.17 (0.17)	0.06 (0.17)	0.11 (0.10)	0.02 (0.20)	-0.15 (0.25)	0.13 (0.10)	0.10 (0.22)	0.25 (0.17)	0.02 (0.08)	0.01 (0.10)
Female	0.24* (0.13)	0.27*** (0.08)	0.35* (0.20)	0.23*** (0.07)	0.53*** (0.18)	0.92*** (0.30)	0.24*** (0.08)	0.49** (0.23)	0.41*** (0.12)	0.04 (0.07)	-0.01 (0.07)
Observations	1,620	1,488	834	2,270	484	426	1,698	818	1,072	1,218	828
Panel B. Poisson Model											
Male	0.55*** (0.18)	0.49** (0.21)	0.22 (0.23)	0.38*** (0.14)	0.61* (0.35)	0.83* (0.47)	0.36** (0.14)	0.52* (0.31)	0.63*** (0.24)	0.16 (0.15)	0.12 (0.17)
Female	1.01*** (0.21)	0.61*** (0.11)	0.69*** (0.22)	0.70*** (0.11)	1.06*** (0.25)	1.23*** (0.34)	0.51*** (0.13)	1.16*** (0.25)	0.90*** (0.17)	0.46*** (0.13)	0.32*** (0.15)
Observations	2,653	2,501	1,344	3,800	811	707	2,812	1,341	1,765	2,048	1,409

Note: This table presents estimates from an over-identified model where treatment indicators (interacted with sex) instrument for $\log(1 - \kappa_{it})$. All models control for the strata used for random assignment, date fixed effects, and for baseline covariates, interacted with a female dummy. Panel A presents scaled estimates from two-stage least squares regression models estimated using the weeks of “fee-free driving”. We scale the coefficients by the control participation rates in each sex-specific subgroup so that we recover elasticities, rather than semi-elasticities. Panel B presents estimates from an IV-Poisson model. In these specifications we use data from both the fee-free driving weeks and from the two “taxi” weeks. Samples are constructed using data from non-treatment weeks. “Late-night” drivers complete >40% of their weekly hours between the hours of 10 PM and 4 AM. “Weekend” drivers complete >50% of their weekly hours on the weekend. “School pick-up” drivers drive for at least 20 minutes between 2 PM and 5 PM on >50% of the weekdays they drive. Standard errors are clustered by driver. Levels of significance: * 10%, ** 5%, and *** 1%.

Table A10: Shifters vs. Non-Shifters

	All Experiments				Houston Only		
	Non-Shifter Mean (1)	Shifter Mean (2)	Raw Difference (3)	Adjusted Difference (4)	Houston 1 Mean (5)	Houston 2 Mean (6)	Difference (7)
Female	0.48	0.30	-0.18*** (0.013)	0.00 (0.027)	0.52	0.41	-0.11*** (0.016)
Months on Platform	15.48	9.54	-5.94*** (0.268)	1.33** (0.574)	16.02	8.31	-7.72*** (0.307)
Vehicle Solutions	0.11	0.08	-0.03*** (0.008)	0.09*** (0.008)	0.12	0.07	-0.05*** (0.009)
Vehicle Year	2012.55	2012.81	0.26** (0.102)	— [†]	2013.55	2013.61	0.06 (0.083)
Prior Week Hours	17.86	14.34	-3.52*** (0.453)	0.97 (0.962)	18.33	14.58	-3.75*** (0.536)
Prior Week Earnings	239.69	220.11	-19.59*** (6.433)	17.63 (17.897)	240.90	202.62	-38.28*** (7.247)
Joint F-statistic			167.50				145.85
<i>p</i> -value			0.000				0.000
Drivers	2,171	3,409			1,976	2,004	

Note: This table compares observable characteristics of shifters (Houston 2 and Boston drivers eligible for Lyft) to non-shifters (Houston 1 and Boston drivers ineligible for Lyft). Columns 1–2 report group means across all experiments. Column 3 reports the raw difference; Column 4 adjusts for experiment fixed effects (because Houston 1 is entirely non-shifters and Houston 2 entirely shifters, Column 4 is identified from the Boston sample alone). The experiment-adjusted difference for vehicle year ([†]) is omitted: shifter status in Boston is defined by vehicle-year eligibility for Lyft, so that within-experiment difference is mechanical rather than a test of balance. Columns 5–7 restrict to Houston only (Houston 1 vs. Houston 2). Standard errors are in parentheses. Levels of significance: *10%, ** 5%, and *** 1%.

Table A11: Shifter Experiment Offers and Opt-In Decisions

	Offered $\log(1 - \kappa)$ (1)	$1\{\text{Accepted}\}$ (2)	$\log(1 - \kappa)$ (3)	$\log(W)$ (4)
Treated	0.24*** (0.001)	0.65*** (0.010)	0.16*** (0.002)	0.18*** (0.010)
Observations	4,910	4,910	4,910	3,730
Strata Dummies	✓	✓	✓	✓

Note: This table presents the first stage for the shifter experiment, the analogue of Table A2 for the non-shifters. The sample is drivers with access to Lyft (Houston 2 and Boston drivers eligible for Lyft). Each column regresses an outcome on the treatment indicator, controlling for randomization strata: the offered increase in the log net-of-commission rate (Column 1), an indicator for accepting the offer (Column 2), the realized log net-of-commission rate (Column 3), and log hourly earnings (Column 4). Standard errors clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A12: Shifter–Non-Shifter Gap, Propensity-Score Reweighted

	Propensity Score Quartile				Reweighted
	Q1	Q2	Q3	Q4	Estimate
Non-Shifter (τ_{non})	0.44*** (0.13)	0.67*** (0.22)	0.44 (0.54)	-0.00 (0.41)	0.35*** (0.07)
Shifter (τ_{shift})	0.92*** (0.31)	0.79** (0.36)	1.32*** (0.30)	1.28*** (0.19)	0.75*** (0.12)
$\tau_{\text{shift}} - \tau_{\text{non}}$	0.49 (0.37)	0.12 (0.51)	0.88 (0.61)	1.28** (0.50)	0.40*** (0.13)
Observations	2,063	1,756	1,653	1,704	7,158

Note: We estimate each driver's propensity to be a shifter, $P(\text{shifter} \mid X)$, by logit on baseline characteristics and stratify drivers into propensity-score quartiles (Columns 1–4) following a standard propensity-score stratification approach. The final column reports a separate IV-Poisson estimate on the full sample in which non-shifters are reweighted to match the shifter distribution of X . Elasticities are estimated by IV-Poisson (GMM), the paper's preferred specification, with randomization-strata fixed effects. The gap $\tau_{\text{shift}} - \tau_{\text{non}}$ is positive in every quartile and economically large and significant on average, indicating the shifter–non-shifter difference is not driven by observable composition. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A13: Robustness of the Shifter–Non-Shifter Gap

	Pooled		Male		Female	
	1{Drive}	Log(Hours)	1{Drive}	Log(Hours)	1{Drive}	Log(Hours)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Just-Identified Model</i>						
Non-Shifter	0.25*** (0.06)	0.38*** (0.09)	0.18** (0.09)	0.27* (0.15)	0.30*** (0.07)	0.46*** (0.12)
Shifter	0.20 (0.18)	1.03*** (0.13)	-0.00 (0.23)	1.16*** (0.14)	0.51* (0.28)	0.68*** (0.26)
Shifter – Non-Shifter	-0.05 (0.18)	0.65*** (0.16)	-0.18 (0.25)	0.89*** (0.21)	0.21 (0.29)	0.22 (0.28)
Observations	6,140	7,026	3,227	4,598	2,913	2,428
<i>Strata Only</i>						
Non-Shifter	0.23*** (0.06)	0.39*** (0.10)	0.14 (0.09)	0.30* (0.16)	0.30*** (0.07)	0.47*** (0.12)
Shifter	0.20 (0.17)	1.02*** (0.12)	0.01 (0.23)	1.14*** (0.14)	0.51* (0.28)	0.69*** (0.25)
Shifter – Non-Shifter	-0.03 (0.18)	0.63*** (0.16)	-0.13 (0.24)	0.85*** (0.21)	0.21 (0.29)	0.23 (0.28)
Observations	6,140	7,026	3,227	4,598	2,913	2,428
Houston EA	✓	✓	✓	✓	✓	✓
Boston EA		✓		✓		✓

Note: Shifters are drivers in the second Houston experiment and Boston drivers eligible to drive for Lyft. Columns report shifter and non-shifter elasticities and their difference for all drivers pooled (Columns 1–2), men (Columns 3–4), and women (Columns 5–6). Within each group, the dependent variable is an indicator for positive hours (extensive margin, Houston only, scaled by each group’s baseline participation rate) or the log of hours worked, which also includes Boston. The upper block uses just-identified instruments; the lower block uses over-identified instruments with strata-only controls. Gaps and their standard errors are computed by the delta method. The intensive-margin (hours) gap is positive and significant across specifications; the extensive-margin gap is smaller and imprecise. All models control for the strata used for random assignment; standard errors are clustered by driver. Baseline results are in Table 5. Levels of significance: *10%, ** 5%, and *** 1%.

Table A14: Utilization Rates by Shifter Status

	All (1)	Male (2)	Female (3)	Placebo (4)
Non-Shifter	0.586 [0.147]	0.583 [0.137]	0.590 [0.156]	0.591 [0.141]
Shifter	0.615 [0.190]	0.617 [0.186]	0.611 [0.196]	0.584 [0.151]
<i>p</i> -value (equal)	<.001	<.001	0.018	0.317
Observations	1,776 / 1,850	851 / 1,164	925 / 686	721 / 1,055

Note: This table compares daily utilization rates (time en route to a pickup or with a passenger, divided by time on the app) for non-shifters and shifters in the Houston experiments. Non-shifters are from the Spring 2017 experiment (Lyft absent); shifters are from the Fall 2017 experiment (Lyft present). Column 4 is a placebo: Houston Spring 2017 drivers split at the median vehicle year, with “Non-Shifter” = older cars and “Shifter” = newer cars. Because Lyft was absent, vehicle age should not predict utilization differences. Standard deviations in brackets. The *p*-value tests the null of equal utilization using a regression with standard errors clustered by driver.

Table A15: Placebo Test: Pseudo-Shifters in Houston Spring 2017

	1{Drive} (1)	log(Hours) (2) (3)		IV-Tobit (4)	Poisson (5)
Pseudo Non-Shifter	0.15** (0.07)	0.39*** (0.14)	0.69*** (0.15)	0.95*** (0.28)	0.44*** (0.14)
Pseudo Shifter	0.14*** (0.05)	0.44*** (0.11)	0.69*** (0.11)	0.97*** (0.21)	0.50*** (0.11)
<i>p</i> -value (equal)	[0.921]	[0.790]	[0.985]	[0.936]	[0.752]
Observations	3,108	2,766	4,650	3,108	3,108
Fee-Free Driving Data	✓	✓	✓	✓	✓
Taxi Data			✓		
Strata Dummies	✓	✓	✓	✓	✓
Baseline Covariates	✓	✓	✓	✓	✓

Note: This table splits the Houston 1 sample (Spring 2017, when Lyft was absent) at the median vehicle year (2014). “Pseudo-shifters” have newer vehicles; “pseudo-non-shifters” have older vehicles. Under the null that vehicle age does not independently affect labor supply elasticities, the two groups should have equal elasticities. The *p*-values confirm that pseudo-shifters and pseudo-non-shifters are statistically indistinguishable across all specifications. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A16: Sensitivity of Optimal Markdowns to Market Share and Estimator

	2SLS				Poisson			
Calibration anchor ϕ_0	0.67	0.72	0.83	0.93	0.67	0.72	0.83	0.93
<i>Panel A: Inputs</i>								
τ_{non} (non-shifter elasticity)		0.42				0.47		
Non-shifter markdown		70.2%				67.8%		
$\varepsilon + \sigma $ (calibrated at ϕ_0)	0.79	1.00	1.90	5.17	0.80	1.01	1.92	5.22
<i>Panel B: Shifter Markdown at ϕ, by Calibration Anchor</i>								
$\phi = 0.67$	55.1%	52.1%	42.3%	25.2%	53.5%	50.7%	41.3%	24.7%
$\phi = 0.72$	57.7%	55.1%	46.2%	29.1%	56.0%	53.5%	45.0%	28.5%
$\phi = 0.83$	63.0%	61.4%	55.1%	40.2%	61.0%	59.5%	53.5%	39.3%
$\phi = 0.93$	67.4%	66.7%	63.8%	55.1%	65.1%	64.5%	61.8%	53.5%

Note: This table reports the optimal monopsony markdown $1/(1+\tau)$ under different assumptions about Uber's market share ϕ and across two estimators (2SLS and Poisson; the IV-Tobit version is in Appendix Table A5). Panel A reports the non-shifter (single-firm) elasticity τ_{non} and markdown, which depend on neither ϕ nor the calibration anchor, together with the calibrated composite for each anchor ϕ_0 , backed out from the Houston shifter–non-shifter gap in Table 5 as $(\tau_{\text{shift}} - \tau_{\text{non}}) \phi_0 / (1 - \phi_0)$. In the 2SLS columns the inputs are the intensive (log-hours) estimates, so this composite is exactly $\varepsilon + |\sigma|$ (Equation 3). In the Poisson columns the inputs are the participation-inclusive totals. Panel B reports the shifter (two-firm) markdown $1/(1+\tau(\phi))$, with $\tau(\phi) = \tau_{\text{non}} + ((1-\phi)/\phi)(\varepsilon + |\sigma|)$, evaluated at each benchmark share ϕ (rows) under each calibration anchor ϕ_0 (columns). Diagonal entries ($\phi = \phi_0$) equal the markdown computed directly from the estimated shifter elasticity; off-diagonal entries are counterfactual markdowns at market shares other than the calibration share. The anchors and evaluation shares are the four benchmarks used throughout the paper: symmetric driver tastes ($\phi = 0.67$), Uber's share of the US ride-share market ($\phi = 0.72$; Campbell, 2022), the lower bound implied by the single-homing rate in Aucejo, Perry, and Zafar (2024) ($\phi = 0.83$), and the external calibration of Koustas (2018) ($\phi = 0.93$). All estimates use Houston-only data from Table 5.

Table A17: Houston Driver Characteristics by Sex

	Active Drivers		Eligible		Experimental	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Age	–	–	–	–	44.0 (12.5)	44.3 (11.9)
Vehicle Solutions	0.07	0.10	0.09	0.13	0.09	0.15
Commission pricing	25.76	26.72	26.58	27.22	24.17	26.53
Months on platform	12.7 (10.0)	9.8 (8.6)	10.8 (8.8)	9.2 (8.0)	20.1 (9.5)	12.2 (9.3)
Vehicle year	2013.0 (3.1)	2013.4 (2.6)	2012.9 (3.2)	2013.5 (2.6)	2013.4 (2.5)	2013.7 (2.5)
Hours per week (pre-period)	18.9 (22.7)	10.3 (11.3)	18.5 (14.6)	13.4 (11.2)	20.7 (14.5)	18.6 (12.7)
0–5 hours	0.30	0.44	0.16	0.21	0.11	0.11
5–15 hours	0.28	0.33	0.37	0.46	0.31	0.35
15–25 hours	0.15	0.13	0.20	0.18	0.28	0.26
25–35 hours	0.10	0.06	0.12	0.09	0.12	0.17
35–45 hours	0.07	0.03	0.08	0.04	0.10	0.07
45+ hours	0.10	0.02	0.07	0.02	0.08	0.04
Observations	12,667	3,149	8,529	2,112	972	1,048

Note: This table describes active Houston drivers (Columns 1 and 2), the subset eligible for our non-shifter experiment (Columns 3 and 4), and the experimental sample (Columns 5 and 6). Section 3.2 defines the samples. The hours-per-week rows report pre-period weekly hours. We screened on eligibility using average weekly hours conditional on working in the four weeks before the experiment (5–40 hours); because hours vary from week to week and the bins here are not conditioned in the same way, the distribution includes mass below 5 and above 45.

Table A18: Shifter and Non-Shifter Elasticities Using Fares

	$1\{\text{Earn}\}$	$\text{Log}(\text{Fares})$
	(2SLS)	(2SLS)
	(1)	(2)
Non-Shifter (τ_{non})	0.13*** (0.04)	0.46*** (0.10)
Shifter (τ_{shift})	0.15*** (0.05)	1.24*** (0.14)
$\tau_{\text{shift}} - \tau_{\text{non}}$	0.02 (0.07)	0.78*** (0.17)
p -value (equal)	0.77	0.00
Observations	5,580	4,929

Note: Shifter and non-shifter elasticities estimated with fares rather than hours as the outcome. Column 1 uses an indicator for positive fares in the week (the extensive margin); Column 2 uses log fares among weeks with positive fares (the intensive margin). Because fares accrue only on completed Uber trips, they are measured with minimal error and are unaffected by how the app records online time during multi-apping. Standard errors are clustered by driver. Levels of significance: *10%, ** 5%, and *** 1%.

Table A19: Estimates of Elasticities Computed Using Complier Means

	Complier Hours			Impact & Elasticity		
	Untreated Mean (1)	Treated Mean (2)	% Increase (3)	Impact on $\log(1 - \kappa)$ (4)	Implied τ_{non} or τ_{shift} (5)	Implied σ (6)
Non-Shifter	18.81 (0.59)	21.74 (0.46)	16%	0.30*** (0.002)	0.53	
Shifter	14.50 (0.48)	18.77 (0.39)	29%	0.24*** (0.001)	1.21	8.45
<i>Male</i>						
Non-Shifter	20.47 (1.00)	22.62 (0.71)	11%	0.28*** (0.003)	0.38	
Shifter	15.94 (0.59)	20.47 (0.45)	28%	0.25*** (0.001)	1.16	9.95
<i>Female</i>						
Non-Shifter	17.52 (0.69)	21.12 (0.60)	21%	0.31*** (0.002)	0.67	
Shifter	10.53 (0.74)	14.16 (0.70)	35%	0.24*** (0.003)	1.45	9.68

Note: Columns 1 and 2 present estimates of the treated and untreated complier mean hours worked, following Abadie (2002). Column 3 presents the percent increase in hours worked associated with the treatment. Column 4 presents the impact of the treatment on the log net-of-commission rate. Column 5 presents implied elasticities (τ_{non} for non-shifters, τ_{shift} for shifters), computed by dividing the proportional increase in hours worked (Column 3) by the impact on the log net-of-commission rate (Column 4). Column 6 presents an implied firm-substitution elasticity $|\sigma|$, obtained by applying the decomposition (Equation 3) to the shifter–non-shifter gap in the Column 5 elasticities at the Koustas (2018) benchmark $\phi = 0.93$. The top panel splits drivers by shifter status; the bottom panel further splits by sex. Standard errors are in parentheses.

B Theoretical Appendix

B.1 Proof of Result 1

Setup. Following the framework in Section 2.2, workers have an increasing, strictly convex cost of hours $v_i(H)$ with $v_i(0) = 0$, pay a fixed participation cost $\chi_i > 0$ to work at the inside firm, and have access to an outside employer offering earnings $S_i(r)$, where S_i is increasing and strictly concave with $S_i(0) = 0$. Conditional on participating inside, worker i chooses inside hours h and outside hours r to maximize $wh + S_i(r) - v_i(h+r)$; let $U_i^B(w)$ denote the resulting maximum utility. If she does not participate inside, she earns $U_i^O := \max_r [S_i(r) - v_i(r)]$. She participates inside whenever $U_i^B(w) - U_i^O \geq \chi_i$.

Worker i 's reservation wage b_{i1} is the smallest wage at which she would participate inside: $U_i^B(b_{i1}) - U_i^O = \chi_i$. Without the outside employer ($S_i \equiv 0$), $U_i^O = 0$ and the corresponding reservation wage b_{i0} satisfies $U_i^A(b_{i0}) = \chi_i$, where $U_i^A(w) := \max_h [wh - v_i(h)]$ is the worker- i inside utility. Unlike Section 2.2, costs v_i are worker-specific here; we do not impose the common-cost simplification under which U_i^A reduces to the scalar $u(w)$ defined there. Let F_0 and F_1 denote the CDFs of reservation wages without and with the outside employer, with densities f_0 and f_1 .

Part 1: FOSD. We first show that $U_i^B(w) - U_i^O \leq U_i^A(w)$ for every wage w . Since v_i is convex with $v_i(0) = 0$, $v_i(h+r) \geq v_i(h) + v_i(r)$ for all $h, r \geq 0$. Therefore

$$U_i^B(w) = \max_{h,r} [wh + S_i(r) - v_i(h+r)] \leq \max_h [wh - v_i(h)] + \max_r [S_i(r) - v_i(r)] = U_i^A(w) + U_i^O.$$

Evaluated at the reservation wages, $U_i^A(b_{i0}) = \chi_i = U_i^B(b_{i1}) - U_i^O \leq U_i^A(b_{i1})$. Because U_i^A is increasing in w , $b_{i1} \geq b_{i0}$. The rise is strict whenever the worker's outside-only utility is positive ($U_i^O > 0$): if she sets $r^* = 0$ at b_{i1} , then $U_i^B(b_{i1}) = U_i^A(b_{i1})$, so $U_i^A(b_{i1}) = \chi_i + U_i^O > \chi_i = U_i^A(b_{i0})$; if instead $r^* > 0$, she also sets $h > 0$ (participation must cover $\chi_i > 0$), and strict convexity of v_i makes the superadditivity inequality strict, so again $U_i^A(b_{i1}) > \chi_i$. Each worker's reservation wage rises weakly, so for any b the set

$\{i : b_{i1} \leq b\} \subseteq \{i : b_{i0} \leq b\}$, and $F_1(b) \leq F_0(b)$ for all b .

Part 2: MLRP implies η^{firm} rises. The firm-specific elasticity is $\eta^{\text{firm}}(w) = wf(w)/F(w)$, which is proportional to the reverse hazard rate $\tilde{r}(w) := f(w)/F(w)$. We show that if $f_1(b)/f_0(b)$ is nondecreasing in b (MLRP, with the convention that the ratio equals zero where $f_1(b) = 0 < f_0(b)$), then $\tilde{r}_1(w) \geq \tilde{r}_0(w)$ at every wage w with $f_1(w) > 0$ and $F_1(w) > 0$, with strict inequality whenever the likelihood ratio is not constant below w .

Fix such a w and split the region below it into $A = \{b \leq w : f_1(b) = 0\}$ and $B = \{b \leq w : f_1(b) > 0\}$. Then

$$F_0(w) = \int_A f_0(b) db + \int_B \frac{f_0(b)}{f_1(b)} f_1(b) db.$$

Because f_1/f_0 is nondecreasing, $f_0(b)/f_1(b) \geq f_0(w)/f_1(w)$ for all $b \in B$. Therefore

$$F_0(w) \geq \int_A f_0(b) db + \frac{f_0(w)}{f_1(w)} \int_B f_1(b) db \geq \frac{f_0(w)}{f_1(w)} F_1(w),$$

where the last step uses $\int_B f_1(b) db = F_1(w)$ (since $f_1 = 0$ on A) and discards the nonnegative first term. Rearranging: $f_1(w)/F_1(w) \geq f_0(w)/F_0(w)$, i.e., $\tilde{r}_1(w) \geq \tilde{r}_0(w)$. If the likelihood ratio is not constant below w , at least one inequality is strict: either A carries positive f_0 -mass (there the ratio is zero, while it is positive at w), or $f_0(b)/f_1(b) > f_0(w)/f_1(w)$ on a positive-measure subset of B . In either case $\tilde{r}_1(w) > \tilde{r}_0(w)$, and since $\eta^{\text{firm}}(w) = w \cdot \tilde{r}(w)$, the firm-specific elasticity strictly rises. In the example of Figure 1, f_0 is uniform and the outside option raises the floor of the support; the likelihood ratio is zero below the new floor and constant above it, so the strict increase comes from the excluded region A . $\square$

When MLRP is plausible. The intuition for MLRP is that workers with weaker existing alternatives receive larger gains from access to the new employer, so the outside option compresses the reservation-wage distribution from below.

There are two caveats. First, with heterogeneous cost functions v_i , participation costs χ_i , and outside-earnings functions S_i , the new reservation wage b_{i1} depends on more than the old reservation wage b_{i0} and the ranking of reservation wages may differ once the new

employer enters. Second, even when the shift is systematically larger at lower b_{i0} and the map is order-preserving, the induced density ratio $f_1(b)/f_0(b)$ also depends on the local rate at which that map stretches the distribution.

B.2 Proof of Result 2

Worker-level result (unconditional). Consider a worker with positive hours at both firms. Her total hours satisfy $w = v'_i(H)$, the same first-order condition as under a single firm. Total hours H and the market-hours elasticity ε_i are therefore unchanged by the outside firm. Define $\phi_i = h_i/H_i$ and let σ_i denote the elasticity of outside hours to the inside wage ($\sigma_i < 0$ by strict concavity of S_i). Since $h = H - r$, the elasticity of inside hours to the inside wage decomposes as

$$\begin{aligned} \frac{\partial h}{\partial w} \frac{w}{h} &= \frac{\partial H}{\partial w} \frac{w}{H} - \frac{\partial r}{\partial w} \frac{w}{r} \\ &= \frac{\partial H}{\partial w} \frac{w}{H\phi} - \frac{\partial r}{\partial w} \frac{w}{H\phi} \\ &= \frac{\partial H}{\partial w} \frac{w}{H} \times \frac{1}{\phi} - \frac{\partial r}{\partial w} \frac{w}{r} \times \frac{1-\phi}{\phi} \\ &= \frac{1}{\phi} \varepsilon_i - \frac{1-\phi}{\phi} \sigma_i. \end{aligned}$$

Since $\phi_i < 1$ and $\sigma_i < 0$, her inside-hours elasticity exceeds ε_i by $(1-\phi_i)/\phi_i \cdot (\varepsilon_i - \sigma_i) > 0$.

Aggregate result (under MLRP and common ε). Under common ε , every participating worker has inside-hours elasticity $\geq \varepsilon$, with strict inequality for those with positive outside hours. The aggregate intensive-margin elasticity is the inside-hours-weighted average of individual inside-hours elasticities, so it weakly exceeds ε . Without the outside firm, $\phi_i = 1$ for all workers and the aggregate intensive-margin elasticity equals ε . Therefore the intensive margin rises, strictly so when a positive mass of participating workers supplies positive outside hours. Under MLRP, the extensive margin also rises: $\eta_1^{\text{firm}} \geq \eta_0^{\text{firm}}$, strictly when the likelihood ratio is not constant below the wage (Result 1). The worker-level additive decomposition $\tau = \eta^{\text{firm}} + \text{aggregate intensive margin}$ is exact when hours conditional

on working are homogeneous across workers. With heterogeneous ϕ_i , two simplifications become approximate: representing the aggregate intensive margin with a single pair (ϕ, σ) , and treating the extensive contribution as the headcount elasticity η^{firm} . The latter is approximate because the true aggregate extensive contribution weights marginal entrants by their hours relative to incumbents' hours. When hours conditional on working differ across workers, both components increase and $\tau_1 > \tau_0$ at every wage at which either strictness condition holds. $\square$

B.3 Relationship Between Firm and Market Participation Elasticities

Section 2 decomposed the total firm-specific elasticity into extensive and intensive margins. Here we derive the analogous decomposition for participation alone, in a setting where workers can work for only one firm at a time.

Let p^{inside} and p^{outside} denote the probabilities of working at the inside and outside firms, $P = p^{\text{inside}} + p^{\text{outside}}$ the overall participation rate, and $\phi := p^{\text{inside}}/P$ the inside firm's share of participants. The overall participation elasticity decomposes as

$$\begin{aligned} \frac{\partial P}{\partial w} \frac{w}{P} &= \frac{\partial p^{\text{inside}}}{\partial w} \frac{w}{P} + \frac{\partial p^{\text{outside}}}{\partial w} \frac{w}{P} \\ &= \frac{\partial p^{\text{inside}}}{\partial w} \frac{w}{p^{\text{inside}}} \times \phi + (1 - \phi) \frac{\partial p^{\text{outside}}}{\partial w} \frac{w}{p^{\text{outside}}}. \end{aligned} \quad (6)$$

The overall participation elasticity is a share-weighted combination of two terms: the inside firm's own participation elasticity (first term, positive—a higher inside wage draws new workers in) and the cross-elasticity of outside participation with respect to the inside wage (second term, negative—a higher inside wage pulls workers away from the outside firm). Rearranging, the inside firm's own participation elasticity equals $1/\phi$ times the overall participation elasticity minus $(1 - \phi)/\phi$ times the (negative) cross-elasticity: as on the hours margin, part of the firm's participation response reflects workers drawn from the outside firm rather than into the market, and the amplification grows as the inside share ϕ falls.

C Empirical Appendix

C.1 Variable Definitions

Hours. We count a driver as working whenever the Uber app is on and the driver has indicated availability for dispatch.

Utilization. Drivers' hours can be split into three distinct periods: time waiting for a trip, time traveling to a pickup, and time on a trip. The utilization rate is the fraction of hours spent in the latter two periods.

Earnings. Uber distinguishes between gross earnings—which include Uber's own platform promotions—and net earnings—which subtract the amount the driver paid in Uber fees. Gross earnings do not include our experimental incentives. Gross earnings and fares are constructed the same way across all our experiments and record only Uber's standard pay, so our increase—whether implemented as a reduction in the Uber fee (the fee-based experiments) or as a direct top-up (the second Houston experiment)—is excluded from them; gross earnings and fares are thus mechanically insulated from treatment in every experiment. Neither measure deducts costs the driver may incur, including gas or vehicle depreciation. We focus on gross earnings. If driver costs are roughly proportional to hours driven, this substitution is without loss of generality.

C.2 The Earnings Accelerator

This section provides additional detail on the three Earnings Accelerator experiments.

C.2.1 Houston: Spring 2017 (Non-Shifter Experiment)

In spring 2017 we conducted an Earnings Accelerator experiment in Houston, Texas. Uber launched operations in Houston in July 2013 and by the spring of 2017 had over 15,000 active drivers (drivers who had completed at least four trips in the previous month). Lyft

entered in February 2014 but suspended operations in November 2016 after the Houston City Council required fingerprint background checks for ride-share drivers. Lyft returned in May 2017 when the Texas state legislature passed statewide legislation (H.B. 100) allowing name-based background checks. Uber remained operational throughout.

There were three phases: (1) the selection of eligible drivers, (2) “fee-free” offers, and (3) taxi offers. Drivers were eligible for inclusion in the experiment if they completed at least four trips in the prior month (i.e., were “active” drivers) and if their average hours per week, conditional on driving, were between 5 and 40 hours per week in the month before the experiment. Within this sample of eligible drivers, we randomly selected 2,020 drivers for inclusion in the experiment within six strata defined by the interaction of hours bandwidth and sex. Houston drivers faced either a 20% or 28% commission. “Low-hours” drivers drove an average of 5–15 hours per week; “high-hours” drivers drove an average of 15–25 hours per week; “very-high-hours” drivers drove an average of 25–40 hours per week. We oversampled female drivers so that we would have roughly equal proportions of male and female drivers within each hours stratum.

We offered half of the drivers in the experiment the opportunity to drive fee-free in one week, and the other half the same opportunity the next week. Column 3 of Table A3 shows that drivers offered fee-free driving in wave 1 were statistically indistinguishable from those offered fee-free driving in wave 2. We notified drivers about the Earnings Accelerator offer via e-mails, text messages, and in-app notifications. We crafted the messaging to mimic standard Uber promotions. The in-app notification stayed at the top of each driver’s Uber app for the entire opt-in period, and drivers received reminder e-mails and text messages throughout. Each message contained a link to a Google Form, which provided more information on the incentive. The form indicated the exact time the incentive would be active (Monday 4 a.m. for one week, following the standard Uber week) and informed drivers that if they opted in to the Earnings Accelerator, their data would be used by academic researchers. In all, 1,355 drivers accepted our offer, and we included them in the third phase of the experiment.

At the time we ran the experiment, drivers' trip receipts typically showed three things: the amount collected from the rider, the amount collected by Uber (due to the proportional fee), and the amount they were paid (the difference between the two; see Figure A1 for a sample trip receipt). Drivers who accepted the offer of fee-free driving could see in-app that their fees were zero. They also received e-mail, text message, and in-app reminders throughout the week that the "Earnings Accelerator [was] on" and that they were earning more on every trip.

The third phase of the experiment consisted of two weeks of taxi offers. Figure A11 shows the sliders provided to drivers in the taxi treatments to help them gauge trade-offs. Table A20 shows the taxi contracts offered in each of the two taxi weeks, along with the probability of selection and the percentage of drivers who accepted our offers. Each taxi contract provided the driver with fee-free driving in exchange for a one-time payment (listed in Columns 1 and 5). Columns 3 and 4 of Table A3 show that both phases of the non-shifter experiment were balanced.

C.2.2 Houston: Fall 2017

We conducted a second Earnings Accelerator experiment in Houston, several months after Lyft re-entered the market. On May 17, 2017, the Texas state legislature passed H.B. 100, with a supermajority in the Senate (21–9); on May 29, the Governor signed it, immediately removing mandatory fingerprinting. Lyft announced its intention to resume operations and re-entered Houston at 2 p.m. CT on May 31, 2017.

Drivers were eligible for inclusion in the second Houston Earnings Accelerator experiment if—as before—they had completed at least four trips in the prior month (i.e., were "active" drivers), if their average hours per week, conditional on driving, were between 5 and 40 hours per week, and if they had completed a trip in Houston after Uber resumed operations following Hurricane Harvey.³⁵ The messaging, notifications, and timeline were

³⁵We delayed the start of the experiment until the total volume of trips had rebounded to pre-hurricane levels. We did not want to include drivers who had stopped driving because they had been personally affected by the hurricane.

similar to those in the first Houston experiment. The one key change was reference to Uber’s fee: Uber changed its policy in June 2017 to loosen the link between rider fares and driver earnings—known as “upfront pricing”—and removed the concept of the Uber “fee.” What drivers earned per trip did not change; it remained a function of a base fare plus a per-mile rate and a per-minute rate. As a result, we did not mention the “Uber fee” in the second Houston experiment. We instead focused our messaging on the proportional increase in earnings. Column 6 of Table A3 shows that treatment and control drivers were balanced on observables.

We included 2,100 drivers in the second Houston experiment, for which we have one week of opt-in data and one week of treatment data. During opt-in week (the week of September 18), treatment group drivers were offered one of four multipliers on total earnings at no cost (the equivalent of fee-free driving): 1.2x, 1.3x, 1.4x, and 1.5x. Those who opted in saw the treatment in-app (as a proportional increase in the base fare, per-mile rate, and per-minute rate) the following week. We delivered the treatment as in the other two Earnings Accelerator experiments; it was visible on drivers’ apps in real time.

After the first week of the second Houston experiment was completed, Uber changed (without the authors’ prior knowledge) the driver app, making it impossible for us to deliver treatments in a way that allowed drivers to see or track their higher earnings in real time. Instead, drivers saw only their non-treated earnings until the treatment was over. The change affected only our ability to show Earnings Accelerator treatment payments in real time; it did not affect drivers’ ability to see, e.g., Surge in real time. These weeks also coincided with a change in the Uber app—which very likely affected drivers’ behavior absent treatment. Because we have no experiment that compares the impact of these ex-post payments pre- and post-Lyft (and the treatments we were able to deliver after the app change were not comparable to those in the Boston or first Houston experiments), we do not include data from weeks after the app change. We view this as a conservative choice. A previous draft documented similar results when these weeks were included.

C.2.3 Boston Experiment

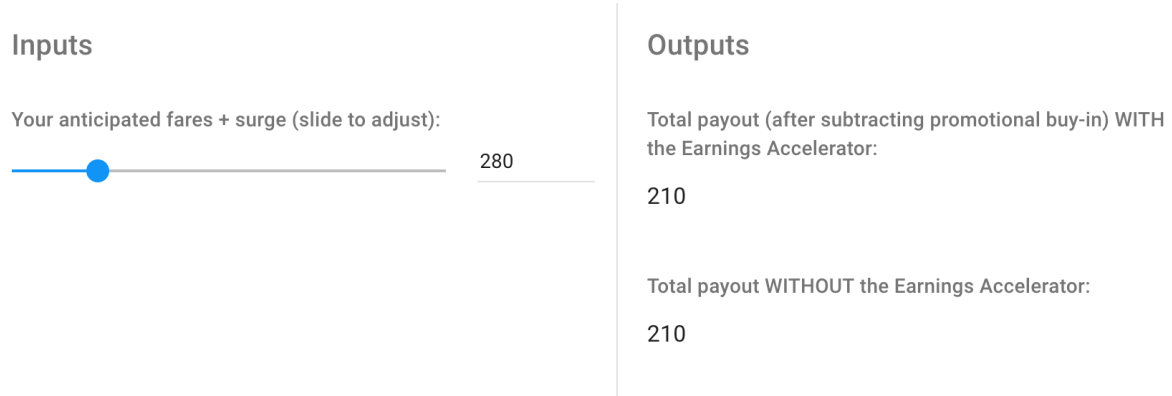
The experiments we conducted in Houston were based on an Earnings Accelerator experiment conducted by Angrist, Caldwell, and Hall (2021) in August–October 2016. As in the first Houston experiment, there were three phases: (1) the selection of eligible drivers, (2) “fee-free” offers, and (3) taxi offers. Because the taxi offers differed from those used in Houston, we include data only from the two weeks of “fee-free driving” offers.³⁶ These offers were comparable to those in both Houston experiments.

Drivers were eligible for inclusion in the Boston experiment if they had completed at least four trips in July 2016 (i.e., were “active” drivers) and if their average hours per week, conditional on driving, were between 5 and 25 hours per week. Drivers were grouped into two bandwidths based on their average hours per week. “Low-hours” drivers drove an average of 5–15 hours/week and “high-hours” drivers drove an average of 15–25 hours/week. Unlike the Houston experiments, the Boston experiment had no “very high” hours group.

We selected 1,600 eligible drivers for the experiment, within strata defined by (1) average hours driven in July, (2) driver fee class (commission rate), and (3) vehicle model year. All 1,600 of these drivers were offered one week of fee-free driving, half in one week (wave 1) and half in the next (wave 2). Column 2 of Table A3 shows that drivers offered fee-free driving in wave 1 were statistically indistinguishable from those offered fee-free driving in wave 2. The lower proportion of female drivers in Boston reflects that Angrist, Caldwell, and Hall (2021) did not oversample female drivers.

³⁶One concern is that the taxi offers are attractive only to high-hours drivers. As a result, the set of compliers depends on the precise parameters of the taxi offer. We find that elasticities differ by usual hours worked.

Figure A11: Earnings Accelerator Taxi Offer



Note: Each driver who was offered a taxi contract was sent a slider that allowed them to compare the earnings they would receive if they accepted the offer (net of the lease) to the earnings they would normally receive. The slider was set to load at the breakeven (the place where treated and untreated earnings would be identical).

Table A20: Taxi Treatments and Opt-In

Bandwidth	20% Fee Class			28% Fee Class		
	Lease	Treatment Fraction	Opt-In Rate	Lease	Treatment Fraction	Opt-In Rate
(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Week 1</i>						
Very High	\$100	60%	36%	\$120	60%	36%
High	\$40	60%	47%	\$50	60%	46%
Low	\$15	60%	47%	\$15	60%	52%
<i>Week 2</i>						
Very High	\$65	60%	47%	\$90	60%	39%
High	\$35	60%	45%	\$35	60%	46%
Low	\$10	60%	57%	\$10	60%	52%

Note: This table presents the taxi treatments offered to drivers included in the taxi phase of the main (pre-Lyft) Earnings Accelerator experiment. Only drivers who accepted fee-free driving were included. The treatment fraction refers to the fraction of consented drivers in each hours bandwidth and commission who were offered a given taxi offer. The opt-in fraction is the fraction of drivers offered the treatment who accepted it.

D Additional Results and Discussion

D.1 Driver Opt-In Decisions

The offer of fee-free driving was generous, both in absolute terms and relative to standard Uber promotions, yet not everyone accepted. Conversations with local city teams suggest that drivers sometimes ignore Uber’s promotional messaging.

Figure A6 plots the opt-in rate by the number of hours a driver worked during the week in which they received the treatment offers (“opt-in week”). For both male and female drivers, those who drove during opt-in week were about 30 percentage points more likely to accept our offer. One delivery channel was the in-app message, which drivers would not see unless they opened the app during opt-in week.

Conditional on positive hours, opt-in does not vary much with hours worked, which is consistent with drivers responding to the proportional rather than the total payoff (though a generous offer that clears the opt-in cost even for low-hours drivers could also produce a flat profile). The higher opt-in among drivers active in opt-in week is consistent with their seeing the in-app messaging, but those drivers are also more engaged and more likely to have near-term driving plans, and we cannot observe which channel led to acceptance.

We do not have data on which link (the e-mail, text message, or Uber app) a driver clicked to access the offer acceptance form.

D.2 First Stage Impact on Hourly Earnings and the Net-of-Commission Rate

Drivers’ hourly earnings depend on both what they collect in trip receipts and the commission they face. Our treatment increased drivers’ hourly earnings by giving them the opportunity to face a lower commission on each trip or by giving them a proportional “bonus” on each trip. This is analogous to lowering the tax rate on drivers’ income. Drivers’ realized hourly earnings, W , depend on both their hourly trip receipts w and the commission they

face, κ : $W = (1 - \kappa) \times w$. Drivers' hourly earnings can change due to either changes in the commission rate or changes in hourly trip receipts:

$$\Delta \log W = \Delta \log(1 - \kappa) + \Delta \log w$$

Our treatment targets per-trip earnings, but average hourly earnings may change if marginal hours are less profitable than average hours.

The impact on wages depends on both the effect of the treatment on treated drivers and experimental take-up. Column 3 of Panel B of Table A2 shows that roughly 60% of male drivers accepted the offer, relative to roughly 70% of female drivers. These opt-in rates are high relative to those for typical Uber promotions.

Female drivers in our sample faced higher commissions on average because they had joined the platform more recently, so they received more generous offers on average. Column 1 of Panel B of Table A2 shows that the average female driver was offered a 0.31 log-point increase in the net-of-commission rate, compared with 0.28 for male drivers. Table A21 shows that differences in treatment generosity do not explain sex differences in opt-in rates; there is no within-sex difference in opt-in rates between drivers who faced different commission rates.

The impact on the log net-of-commission rate is the average offer multiplied by the experimental opt-in rate. Column 5 of Panel B of Table A2 shows that the average impact on the log net-of-commission rate was 0.22 for female drivers, relative to 0.17 for male drivers. The impact on log hourly earnings (Column 7 of Panel B of Table A2) is somewhat smaller than the impact on the net-of-commission rate—the difference is statistically significant but economically modest—consistent with marginal hours being somewhat less productive. Because the hourly-earnings impact is estimated on driver-weeks with positive hours while the net-of-commission impact uses all driver-weeks, we treat the implied shortfall as approximate. (The marginal-hour productivity decline it implies is larger, since it is concentrated on the added hours.)

Table A21: Earnings Accelerator Opt-In Rates by Sex and Commission

	Full Sample (1)	20% Commission (2)	28% Commission (3)
Male			
Treated	0.61*** (0.016)	0.61*** (0.023)	0.61*** (0.022)
Observations	1,944	930	1,014
Female			
Treated	0.73*** (0.014)	0.76*** (0.031)	0.72*** (0.015)
Observations	2,096	386	1,710

Note: This table presents opt-in rates for the main (pre-Lyft) Houston experiment, separately by driver sex and commission rate. Within each sample we regress an indicator for whether the driver accepted the offer on an indicator for whether the driver was treated (i.e., given an offer) and on the strata used for random assignment. Each cell presents the coefficient on treatment, the standard errors (clustered by driver) in parentheses, and the sample size. The p -value for equality between the opt-in rates for male (female) drivers of different commission rates is 0.93 (0.22). Levels of significance: *10%, ** 5%, and *** 1%.

D.3 Heterogeneity in Market-Level Elasticities

Table A9 presents elasticity estimates for subgroups defined by observable characteristics. The Poisson estimates (Panel B) additionally draw on the two “taxi” experiments to boost power; the extensive-margin estimates (Panel A) use the fee-free weeks only.

Age, Experience, and Wealth. Elasticities vary little by age, and although female drivers with more experience on Uber are somewhat more elastic than those with less (while male elasticities are flatter), the gap between men and women persists within every subgroup (Columns 1–4 of Table A9): female drivers are more elastic than their male counterparts among younger and older drivers, and among more- and less-experienced drivers alike. Differences in age or experience therefore do not explain the gap. We have limited information on drivers’ outside income or wealth, but female drivers have newer cars on average,

weighing against the idea that wealth differences explain the gap.³⁷

Hours Worked. Elasticities are generally smaller for higher-hours drivers, and this pattern holds for both men and women (remaining columns of Table A9). Panel A shows that extensive-margin effects are driven by drivers in the lower two strata; there is no impact on whether “very high” hours drivers start driving. Within each stratum, female drivers are more elastic. That lower-hours drivers are more elastic is consistent with Mas and Pallais (2019).

We conducted this experiment when Lyft was out of the market, so substitution to Lyft could not drive the heterogeneity. Drivers may combine driving with non-gig employment, but this is unlikely to explain the sex differences because female Uber drivers would need to be substantially more likely to have outside market employment.³⁸ The final column of Table A9 focuses on drivers observed working at least 40 hours a week; these drivers are similarly elastic to those in the overall “very high hours” stratum.

D.4 Average Elasticities for Male and Female Drivers

We document that male and female drivers with similar characteristics have different elasticities. Male and female drivers also differ in average characteristics. Because we designed our experiment to minimize differences in male and female hours worked, our estimates may not generalize to the average male or female driver in Houston.

To shed light on this question, we run specifications that allow the male and female elasticities ε^g to be a function of observable driver characteristics C_i . We then calculate

$$\varepsilon^p := \int \varepsilon^g(C_i) f^p(C_i) dC_i$$

³⁷In results not reported, we examined heterogeneity in wealth as proxied by income in a driver’s zip code and found little heterogeneity.

³⁸If men are more likely than women to combine gig work with other market-based work, this would mean that we overstate elasticities for male drivers and therefore understate the gap. The descriptive statistics on Houston drivers presented in Appendix D.5 suggest that, if anything, female drivers have less outside work.

for different populations p . To allow for both intensive- and extensive-margin adjustments, we use the IV-Poisson model described earlier in the text, which models the level of hours worked and so retains drivers with zero hours. We estimate ε as a function of discrete characteristics C_i using fully interacted models.

We focus on two key characteristics—hours worked and experience—because these are the main differences between male and female drivers in our sample and between drivers in our sample and the full pool of “active” drivers (Appendix Table A17). We use the hours and experience distribution among the RCT sample to group drivers into hours and experience quantiles.

We first reweight ε^f and ε^m to account for the sex-specific distribution of hours worked—the distribution of hours worked in our sample does not match the distribution of the overall population. The top panel of Figure A12 shows that reweighting elasticities to account for heterogeneity in hours worked widens the observed gaps between male and female drivers. This follows because female drivers drive fewer hours on average, and elasticities are larger among low-hours drivers (Table A9). The second panel shows that reweighting by experience alone narrows the gap somewhat—men are more experienced on average—but does not close it: the reweighted female elasticity is 0.66, relative to 0.51 for men.³⁹

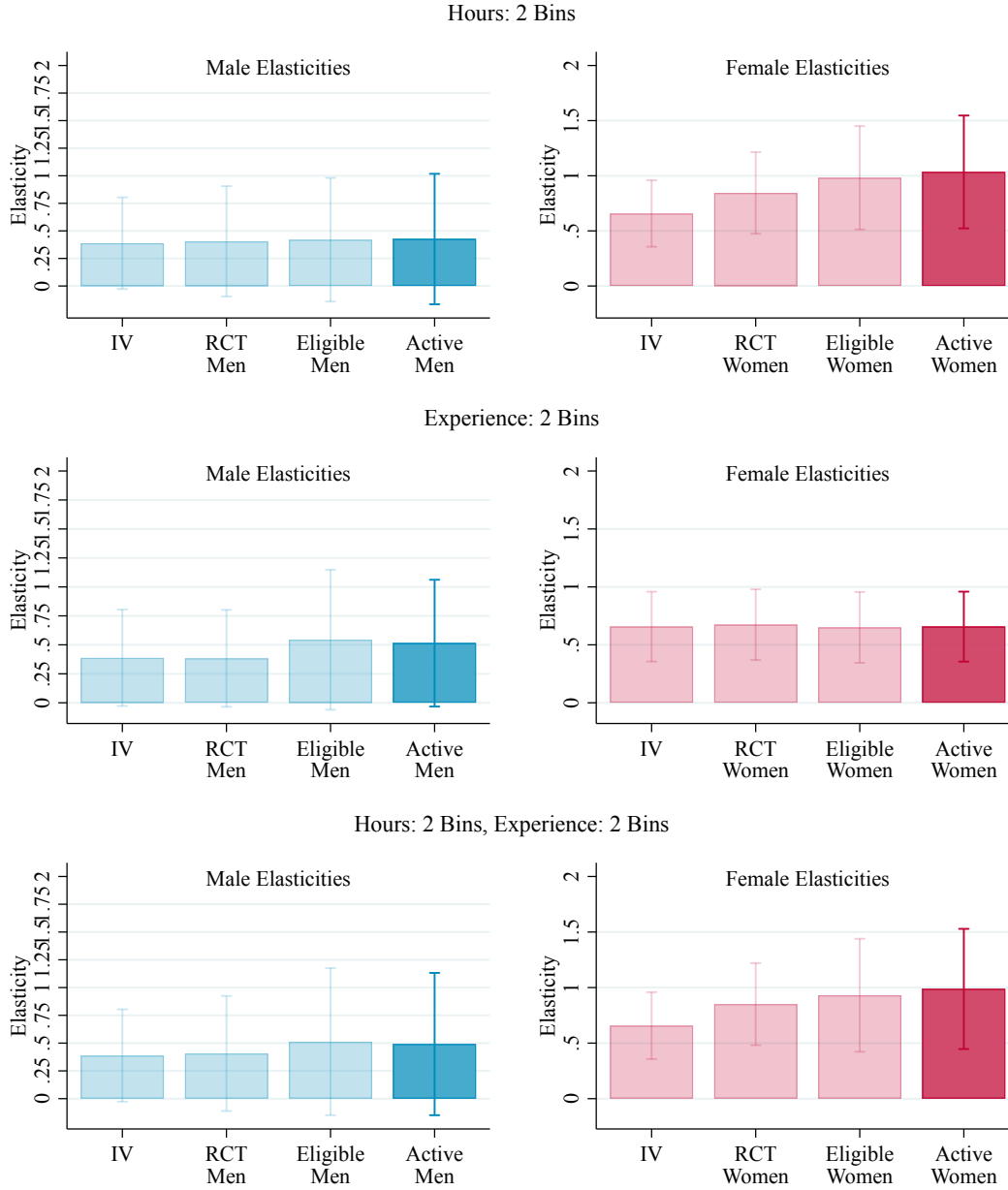
The final panel accounts for differences in both experience and hours. Reweighting to the active driver population does not close the gap between men and women; if anything it widens it. Even after we allow for heterogeneity in both hours and experience, the reweighted female elasticity (0.99) remains larger than the male elasticity (0.49)—a gap of 0.50, slightly wider than the 0.44 gap in the experimental sample (0.85 versus 0.41).

D.5 External Validity and Drivers’ Outside Options

We use data from both the American Community Survey and the Current Population Survey to shed light on whether male and female drivers differ in outside work arrangements.

³⁹We found similar results based on specifications that allow the elasticity to be a more continuous function of these variables.

Figure A12: Elasticities for the Average Driver



Note: We group drivers according to the categories listed in each of the three panels. We then fit an IV-Poisson model of hours worked on $\log(1 - \kappa_{it})$, interacted with group dummies. We use offer-level treatment indicators for each group as instruments and control for strata, interacted with group dummies. We perform this regression separately by driver sex to get estimates of $\varepsilon^f(C_i)$ and $\varepsilon^m(C_i)$. We then calculate the weighted average of these covariate-specific elasticities for the groups indicated. The estimates in the left figure of each panel are reweighted estimates of ε^m ; the estimates in the right figure of each panel are reweighted estimates of ε^f . We define each sample in Section 3.2. Whiskers denote 95% confidence intervals.

Table A22: Complier Characteristics

	Male Drivers		Female Drivers		p-value
	$E[X]$ (1)	$\frac{E[X D_{i1} > D_{i0}]}{E[X]}$ (2)	$E[X]$ (3)	$\frac{E[X D_{i1} > D_{i0}]}{E[X]}$ (4)	
Age	43.84	1.00	44.27	1.02	0.44
Months Since Signup	20.09	0.99	12.22	1.02	<.01
Vehicle Solutions	0.09	1.06	0.15	1.12	<.01
Commission	24.17	1.00	26.53	1.00	<.01
<u>Usual Hours Worked (Pre-Selection)</u>					
Mean	20.67	1.03	18.62	1.05	<.01
Between 0 and 5	0.11	0.70	0.11	0.76	0.75
Between 5 and 15	0.31	1.05	0.35	0.96	0.05
Between 15 and 25	0.28	1.02	0.26	1.04	0.53
Between 25 and 35	0.12	1.11	0.17	1.10	<.01
Between 35 and 45	0.10	0.99	0.07	1.11	<.01
Over 45	0.08	1.03	0.04	1.08	<.01

Note: This table compares the characteristics of fee-free driving compliers to the pool of experimental drivers. Columns 1 and 3 report sample means for male and female drivers. Columns 2 and 4 show how the compliers compare to their same-sex counterparts. Column 5 shows the p -value from a test of equality of the sample means in Columns 1 and 3. We compute complier characteristics following Abadie (2002).

First, in Table A23 we use data from the 2017–2019 American Community Surveys to compare the patterns of hours worked among male and female drivers and male and female taxi drivers in Houston. Columns 1 and 2 use data on the pool of active Houston Uber drivers. Columns 3 and 4 focus on Houston drivers (occupation codes 9110–9149), and Columns 5 and 6 focus on Houston taxi drivers (occupation codes 9140–9149). Among Houston drivers and Houston taxi drivers, there is also a gap in hours worked between men and women. This is true both for means and for the proportion who work more than 35 or 45 hours per week.

We next use data from the 2010–2018 Current Population Surveys to examine the characteristics of male and female taxi drivers in the Houston metro area (Flood et al., 2021). Following Cook et al. (2021), we identify taxi drivers using occupation code 9140. Roughly a quarter of Houston taxi drivers are female, similar to the fraction among Houston Uber drivers.⁴⁰ The average male Houston taxi driver works 39 hours a week. The point estimates for female taxi drivers echo the pattern among ride-share drivers (Table A24)—they work fewer hours (first row), are less likely to work more than 25 hours per week (second row), and work a slightly larger share of their hours at their main job (third row)—though in this smaller sample these individual differences are imprecise.

D.6 Impact on the Distribution of Hours Worked

Our treatment increased drivers’ hourly earnings by reducing the implicit tax rate they face. Panel A of Figure A13 shows that drivers who received the treatment offers drove more than drivers who did not receive the offer. The distribution of hours worked for treated drivers (solid line) lies to the right of that for untreated drivers (dashed line). There is a small change in the probability of driving (the gap between the solid and dashed lines at zero hours). The shift in the hours distribution is larger for female drivers than for male drivers.

Panel B of Figure A13 shows that a similar pattern emerges among the treated and

⁴⁰Uber does not collect data on driver education. The average ride-share driver is more educated than the average worker: roughly a third have college degrees, compared with a quarter among workers in the same markets (Hall and Krueger, 2018).

Table A23: Comparing Active Houston Uber Drivers to Houston Taxi Drivers

	Houston Uber Drivers		Houston Drivers (ACS)		Houston Taxi (ACS)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Age	44.0	44.3	45.0	44.1	43.6	42.5
Vehicle Year	2013.0	2013.4	—	—	—	—
Months on Platform	12.7	9.8	—	—	—	—
<i>Usual Hours Worked</i>						
Mean	18.9	10.3	45.3	36.7	42.1	34.8
Between 0 and 5	0.30	0.44	0.00	0.01	0.00	0.01
Between 5 and 15	0.28	0.33	0.03	0.05	0.08	0.14
Between 15 and 25	0.15	0.13	0.05	0.12	0.06	0.22
Between 25 and 35	0.10	0.06	0.07	0.22	0.11	0.08
Between 35 and 45	0.07	0.03	0.46	0.42	0.44	0.31
Over 45	0.10	0.02	0.40	0.17	0.30	0.24

Note: This table compares the mean characteristics of Houston Uber drivers to drivers in the Houston metropolitan area. We use pooled data from the 2017–2019 American Community Survey. Columns 3 and 4 present characteristics for individuals in driving occupations (occupation codes 9110–9149). Columns 5 and 6 present characteristics for taxi drivers (occupation codes 9140–9149). We compute usual hours worked for Uber drivers using pre-Earnings Accelerator data. Age is observed only for drivers in the experimental sample, so the Uber-driver age in Columns 1–2 is computed over that sub-sample rather than the full active-driver pool.

Table A24: Main Job Hours and Total Hours Among Houston Drivers

	Male Mean	(1)	(2)	(3)	(4)
Usual Hours Worked	38.8	-1.92 (1.92)	-3.06 (1.93)	-2.86 (1.92)	
$1\{\text{Hours} \geq 25\}$	0.90	-0.08 (0.06)	-0.11* (0.06)	-0.11* (0.06)	-0.06 (0.04)
Main Job Hours / Total Hours	0.98	0.010 (0.010)	0.013 (0.010)	0.012 (0.009)	0.005 (0.009)
Education Controls			✓	✓	✓
Marital Status				✓	✓
Usual Hours					✓

Note: This table uses data from the 2010–2018 Current Population Surveys to examine hours worked among Houston metro area (IPUMS metro area code 3362) taxi drivers (IPUMS occupation code 9140). Each row shows the coefficient on a female dummy from a regression of the characteristic indicated in the row on a female dummy and on the characteristics indicated at the bottom of the table. The Male Mean column reports the mean of each characteristic among male drivers; the sample is 24% female. Regressions are weighted using CPS sampling weights. Levels of significance: *10%, ** 5%, and *** 1%.

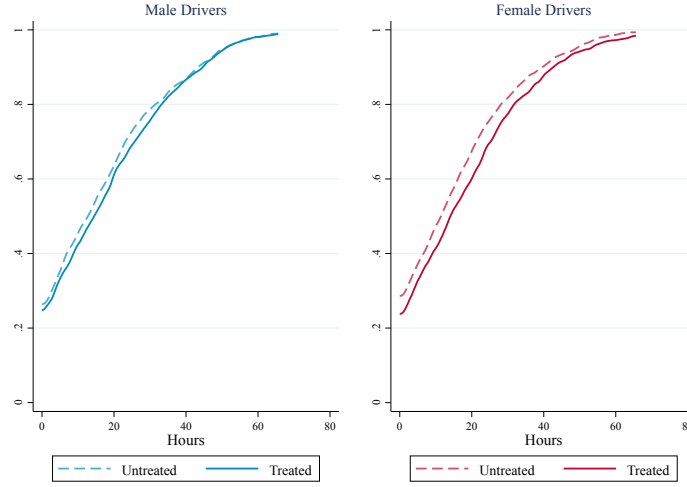
untreated compliers. We estimate complier hours distributions using regressions of the form

$$\begin{aligned}
 1[h_{it} < \nu](1 - D_{it}) &= X'_i\beta_0(\nu) + \alpha_0(\nu)(1 - D_{it}) + u_{0i\nu} \\
 1[h_{it} < \nu]D_{it} &= X'_i\beta_1(\nu) + \alpha_1(\nu)D_{it} + u_{1i\nu},
 \end{aligned}$$

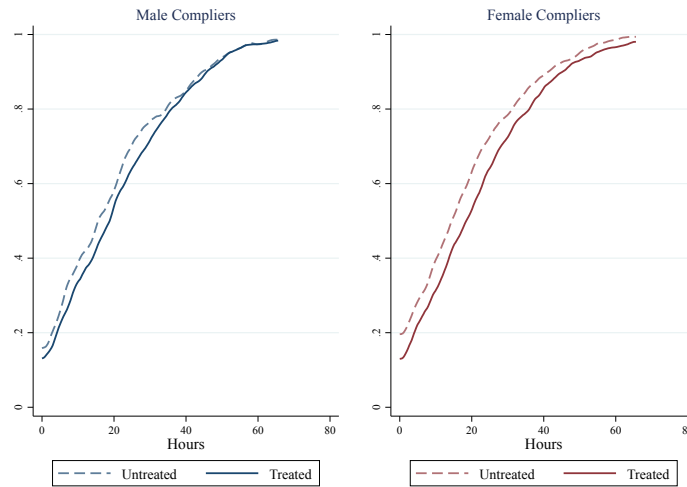
where the outcome variable is an indicator for whether the driver drove fewer than ν hours that week ($1[h_{it} < \nu]$), multiplied by the driver's opt-in decision (D_{it}) or $(1 - D_{it})$ (Abadie, 2002). Because opt-in is endogenous, we instrument for $(1 - D_{it})$ and D_{it} using the randomly assigned treatment offers Z_{it} . Panel B plots estimates of $\alpha_0(\nu)$ and $\alpha_1(\nu)$. Panel B shows that the treatment also shifts the distribution of compliers' hours worked outward. The shift is larger for female drivers than for male drivers.

Figure A13: Effects on the Distribution of Hours Worked

A. All Drivers



B. Compliers



Note: This figure shows the impact of the wage offers on the distribution of hours worked for drivers in the main Houston experiment. Panel A presents the distribution of hours worked for untreated and treated drivers of each sex. Panel B presents the distribution of hours worked for untreated and treated compliers of each sex. The complier distributions are computed using estimates of $\alpha_0(\nu)$ and $\alpha_1(\nu)$ for $\nu \in [0, 80]$ from sex-specific regressions of the form: $1[h_{it} < \nu](1 - D_{it}) = X_i' \beta_0(\nu) + \alpha_0(\nu)(1 - D_{it}) + u_{0i\nu}$ and $1[h_{it} < \nu]D_{it} = X_i' \beta_1(\nu) + \alpha_1(\nu)D_{it} + u_{1i\nu}$ (Abadie, 2002). D_{it} is an indicator for whether driver i accepted an offer in week t . We instrument for $(1 - D_{it})$ and D_{it} using the randomly assigned treatment offers Z_{it} . Each regression controls for the strata used for random assignment.

D.7 Robustness to Potential Measurement Error in Hours

Our hours measure captures the time a driver has the app turned on: waiting for dispatch, traveling to a pickup, and carrying a passenger. Drivers receive no compensation while waiting for a trip. To address potential measurement error in hours, we reproduce our results using drivers' fareboxes. Because this outcome is used to pay drivers, it likely has minimal measurement error. Table A7 shows that we obtain similar results for the market-level elasticities when using log fares instead of log hours as the outcome variable, and Table A18 does the same for the shifter–non-shifter contrast: the intensive-margin outside-option gap is 0.78 in fares, close to its value in hours. Fares carry the additional advantage that they accrue only on completed Uber trips, so they are unaffected by how the app records online time when a driver is running a competing app.

D.8 Impact on When Drivers Work

D.8.1 Hours Worked Before and After Treatment

A natural question is whether treated drivers reduce hours in the week before or after treatment to shift hours into higher-earnings weeks. Appendix Table A8 shows that the treatment offers did not influence drivers' labor supply before or after treatment, consistent with negligible wealth effects. Angrist, Caldwell, and Hall (2021) report similar results based on the Boston Earnings Accelerator experiment. They also document that the treatment did not influence passengers' star ratings, a measure of trip quality.

In unreported results, we used an alternative dependent variable to verify the robustness of the results in Table A8. We obtain virtually identical results when we use drivers' log farebox instead of drivers' log hours as the dependent variable.

D.8.2 Time of Day and Day of Week

Because Uber drivers can drive at any time of day (and any day of the week), we examine where the marginal hours of work come from, separately for male and female drivers. Prior

work by Cook et al. (2021) documents substantial differences in when men and women work. Figure A14 shows the impact of the treatment on whether a driver is active in each hour of the week (Panel A), and how many minutes she works in that hour (Panel B). We run the following two-stage least squares regression separately for each hour-of-week cell:

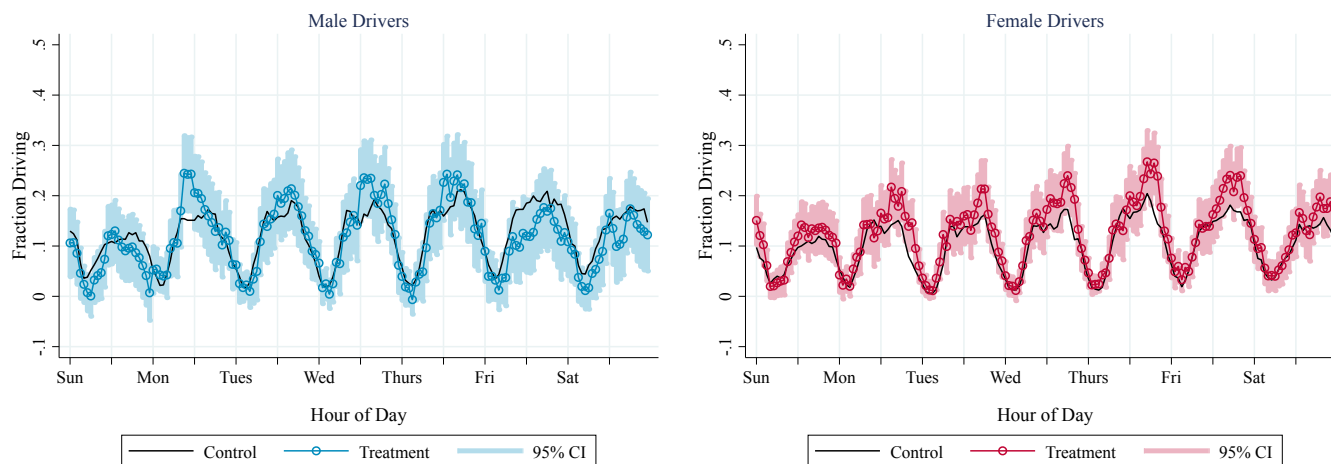
$$\begin{aligned} y_{it} &= \alpha D_{it} + \beta X_{it} + \xi_{it}, \\ D_{it} &= \gamma Z_{it} + \lambda X_{it} + \nu_{it}, \end{aligned} \tag{7}$$

where y_{it} is either an indicator for whether the driver is active or the total number of minutes worked that hour, D_{it} is an indicator for whether the driver accepted the treatment offer, and Z_{it} is an indicator for whether the driver was offered the treatment. The X_{it} include the strata used for random assignment.

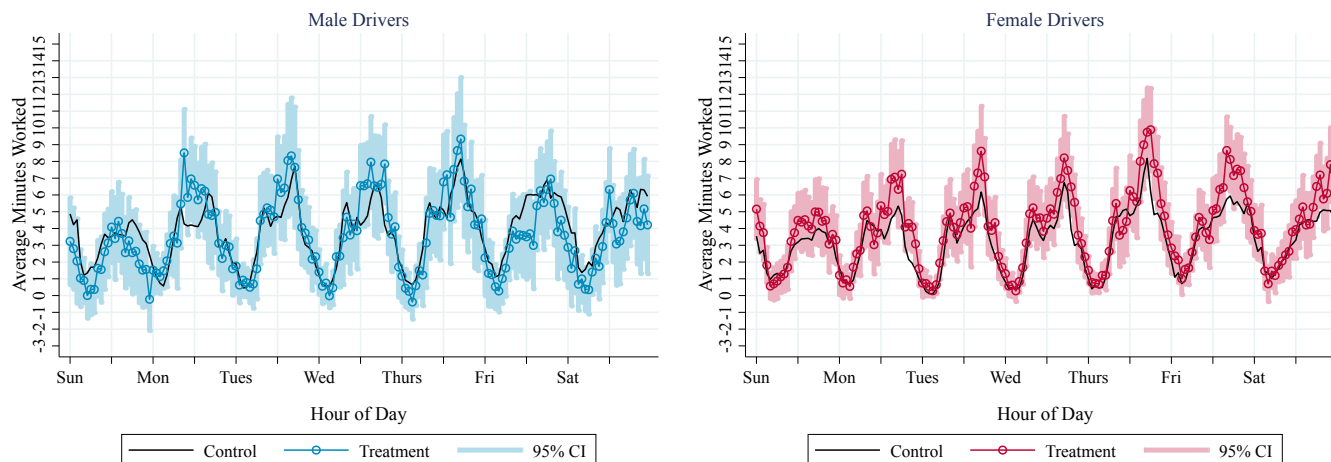
The control means for male and female drivers show a regular pattern: more drivers are active during the day than at night. There is a regular peak at the time most workers end their full-time jobs. Women are less likely to drive late at night. The circles plot the control mean plus the estimated treatment effect (the local average treatment effect of accepting the offer) in each hour-of-week cell; this anchors the effect at the control schedule and is not the mean schedule of treated or complier drivers. The only times when the (shaded) 95% confidence interval does not intersect the control mean are weekday afternoons (especially for female drivers). The 95% confidence intervals intersect the control mean for most outcomes.

Figure A14: Treatment Effects by Time of Day

Panel A: Effect on Driving



Panel B: Effect on Minutes Active



Note: This figure shows the impact of the treatment on when drivers work, plotted across the hours of the week (day-of-week marked on the horizontal axis). The outcomes in Panel A are indicators for whether a driver is driving in a given hour of the week; the outcomes in Panel B are average minutes worked in that hour. The black lines in each figure denote the control means. The dark lines (red/blue) show the control mean plus the estimated treatment effect (anchored at the control schedule). The treatment effects are the two-stage least squares estimates defined above—the local average treatment effect of accepting the offer, instrumented by random assignment. All regressions control for the strata used for random assignment. 95% confidence intervals for each regression are shaded.

D.9 Non-Experimental Estimates and Timing

Our experimental estimates exploit randomly assigned wage variation, eliminating the simultaneity bias that plagues non-experimental estimates of labor supply elasticities. To illustrate this point, Table A4 compares OLS and leave-one-out IV estimates using both non-experimental and experimental weeks of data.

Panel A reports OLS regressions of log hours on log wages. Across all specifications, the OLS estimates are small (0.11–0.23) and similar whether we use non-experimental or experimental weeks, consistent with simultaneity bias attenuating the estimates toward zero. Because log wages are constructed as log earnings divided by hours, division bias—measurement error in hours entering the regressor with the opposite sign—may also attenuate the OLS estimates.

Panel B instruments log wages with the leave-one-out mean of wages among all other drivers in the same week—a standard approach in the non-experimental labor supply literature. Because the instrument varies at the week level, these specifications exclude week fixed effects and draw on across-week variation in market wages. In the non-experimental weeks (Columns 1–3), the IV estimates are near zero or slightly negative. In the experimental weeks (Columns 4–6), the estimates jump to 1.90–2.28 and are highly significant.

Table A8 examines whether our treatment offers influenced drivers’ labor supply in weeks before or after treatment. If drivers anticipated the wage increase and shifted hours intertemporally, we would expect reduced hours in the week before treatment and reduced hours after. Panel A shows that the treatment had no significant effect on the probability of driving in either the pre-treatment or post-treatment period. Panel B shows the same null result for log hours: treated drivers did not reduce their hours before treatment or reduce them after. These results are consistent with the absence of anticipation effects and negligible wealth effects, supporting our market-hours interpretation of the experimental estimates. Angrist, Caldwell, and Hall (2021) report similar findings from the Boston Earnings Accelerator experiment.

E Individual Driver Bonus

E.1 Sample Selection

Section 3.3 describes the IDB program design and randomization. We provide additional details on the sample construction and data structure.

For every week and driver, the data contain up to two observations: one for the weekday period (Monday–Thursday) and one for the weekend period. Roughly 70% of the driver-period observations are for drivers included in the IDB that period. Uber likely did not include drivers in the IDB program at random; drivers included in a given period typically drove more in the previous period. Because we lack information on how Uber selected drivers for the promotion, we cannot investigate external validity as we could in the Earnings Accelerator sample.

Our sample consists of drivers included in the IDB in a given time period. We further:

- Drop the roughly 600 observations where we are missing the age of the vehicle, which prevents us from assigning shifter or non-shifter status ($<0.1\%$ of observations).
- Drop the roughly 21,000 observations for drivers who are older than 70 or younger than 21 ($\approx 2\%$ of observations).

Table A25 shows the fraction of shifters and non-shifters assigned to each trip threshold, as well as the bonus amounts. Non-shifters tended to be assigned to lower thresholds, which carry smaller bonuses.

In some time periods, some strata received two bonus offers: if a driver passed the first trip threshold, she received one bonus, and if she passed the second trip threshold, she received another. Within stratum and period, all drivers received either one offer or two offers. The second threshold was either 70 or 80 trips. For simplicity, we focus on the first (lower) trip threshold and offer.

Table A25: Program Parameters in the Individual Driver Bonus Sample

Threshold	Share of Sample		Bonus (\$)			z_L
	Non-Shifter	Shifter	Mean	Min	Max	
15	0.085	0.066	48.4	15	80	5
18	0.005	0.003	27.5	20	35	5
20	0.194	0.163	63.4	15	130	5
25	0.087	0.073	103.0	60	140	5
30	0.076	0.059	107.8	85	130	10
35	0.094	0.078	82.6	45	145	10
40	0.146	0.140	120.1	90	150	10
50	0.087	0.082	137.3	100	185	10
60	0.059	0.079	195.7	145	245	20
70	0.050	0.092	173.6	110	240	20
75	0.071	0.079	220.5	170	275	20
80	0.046	0.086	151.9	120	195	20
Unique Drivers	684	58244				

Note: For each trip threshold (listed at the left), Columns 1 and 2 list, within the non-shifter (Column 1) and shifter (Column 2) samples, the fraction of driver-period observations assigned to that threshold; each column sums to one across thresholds. The count of distinct drivers in each sample appears in the Unique Drivers row. Column 3 lists the average bonus offered to drivers assigned to that threshold. Columns 4 and 5 list the minimum and maximum bonus offered to drivers assigned to that threshold. Not all thresholds were available in all time periods. Column 6 shows the omitted region used for our bunching estimates. The omitted region was chosen to allow for elasticities up to 1.5 under the maximum bonus associated with each threshold; a few estimated shifter elasticities exceed this value, so for the largest responses the omitted region is modestly conservative and may leave some excess mass in the counterfactual fit.

Table A26: Individual Driver Bonus Balance		
	Control Mean (1)	Difference (2)
Female	0.14	-0.000 (0.001)
Experience (months)	18.75	0.010 (0.027)
Vehicle Year	2013.79	0.001 (0.006)
Lag Trips	30.33	0.021 (0.048)
Lag Earnings (\$)	350.86	-0.107 (0.586)
Previously Treated	0.50	0.000 (0.001)
<i>F</i> -statistic		1.07
<i>p</i> -value		[0.378]
Unique Drivers	55,671 (control), 58,928 (total)	
Observations	513,356 (control), 1,036,336 (total)	

Note: Column 1 shows the mean for drivers in the low bonus treatment. Column 2 shows the strata-adjusted difference between the high and low bonus treatments. None of the drivers in this sample were in the vehicle solutions program. Robust standard errors are in parentheses. Levels of significance: *10%, ** 5%, and *** 1%.

Column 1 of Table A26 shows that IDB drivers are more experienced than those in the Earnings Accelerator samples; the average driver has been on the platform for roughly 19 months, compared with 9 to 16 months across the Earnings Accelerator samples presented in the main text. This reflects that Uber started operations in San Francisco before Houston. Column 2 shows that, conditional on the strata used for random assignment, drivers who received high IDB offers are statistically indistinguishable from those who received low offers.

E.2 Estimating Bunching in the Individual Driver Bonus Sample

This section describes the bunching estimation procedure used to produce the results in Section 5.2. Whether a driver bunches at the threshold depends on both the number of

trips she would have completed in the absence of the incentive and the curvature of her utility function; drivers who are more elastic more readily increase the number of trips they complete in order to receive the bonus. A notch at threshold x of size y leads drivers in an interval $[x - \Delta z^{x,y}, x]$ to bunch at x . The observed amount of bunching— $B_{x,y}$ —depends on both the size of this interval and the shape of the counterfactual trip density $h_0^x(\cdot)$ for drivers assigned to that threshold.

To estimate the amount of bunching for drivers in group g who faced a threshold x and bonus y , we regress the proportion of drivers who completed z trips (p_z^x) on a polynomial of degree D in the number of trips from the threshold, and on a set of dummies for completing trips in the interval $[x - z_L, x + z_U]$ around the threshold:

$$p_z^x = \sum_{d=0}^D \beta_d (z - x)^d + \sum_{j=x-z_L}^{x+z_U} \gamma_j 1\{z = j\} + \xi_z. \quad (8)$$

The lack of g subscripts reflects the fact that we pool the density across all bonus amounts, shifter groups, and non-shifter groups within each threshold–cohort cell. The counterfactual polynomial is therefore common to all groups at a given threshold.

We then calculate the “excess mass” of drivers in group g who faced threshold x and bonus y , as the excess probability of locating at the notch (relative to the polynomial fit). Because this is not comparable across groups with different notches, we divide the excess mass by the counterfactual density at x , $\hat{h}_0^x = \hat{\beta}_0$. Under the standard assumption that the counterfactual density is roughly constant across the omitted region, this scaled excess mass— $\hat{M}_{x,y,g}$ —approximates $\hat{\Delta} z^{x,y,g}$ (Kleven, 2016). Our baseline estimates use a polynomial of degree 3 and exclude 4 bins above the threshold.⁴¹

⁴¹Because we have a wide range of thresholds and bonuses, it would be inappropriate to use the same omitted region for each threshold; low values of z_L may be appropriate for drivers facing low thresholds (e.g., a bonus of \$15 for completing 15 or 20 trips), but would imply implausibly low elasticities for drivers facing higher thresholds and bonuses (e.g., a bonus of \$195 for completing 80 trips). The last column of Table A25 lists the excluded region below the threshold (z_L) for each threshold level.

E.3 Difference in Bunching and Trip Elasticities

We translate bunching estimates into trip elasticities using the structural notch approximation (Kleven and Waseem, 2013; Kleven, 2016). The marginal buncher, whose counterfactual trips are $x - \Delta z^{x,y}$, is indifferent between staying at her counterfactual choice and jumping to the threshold: the bonus must cover the excess effort cost of the additional trips. With per-trip earnings w and a convex cost of trips $c(\cdot)$ satisfying $c'(x - \Delta z^{x,y}) = w$ at her optimum, a second-order expansion gives $y \approx \frac{1}{2} c''(\Delta z^{x,y})^2$; substituting the trip elasticity $\varepsilon^{trip} = w / [(x - \Delta z^{x,y}) c'']$ yields

$$\varepsilon^{trip} \approx \frac{1}{2} \cdot \frac{(\Delta z^{x,y})^2}{(x - \Delta z^{x,y}) \times (y/w)}.$$

The factor $\frac{1}{2}$ reflects that the bonus compensates the average excess cost of the added trips, while the marginal trip costs twice the average. The reduced-form convention, which divides the percent trip change by the average implicit wage change $(y/\Delta z^{x,y})/w$, omits this factor and yields elasticities exactly twice ours.

Table A27 demonstrates that our results are robust to the choice of density estimation parameters, trimming cutoffs, and estimation method. We vary one parameter at a time from our preferred specification: the polynomial degree (Panel A), the width of the counterfactual estimation window (Panel B), the number of excluded bins above the threshold (Panel C), the trimming cutoff for extreme estimates (Panel D), and the estimation method (Panel E). The shifter premium is positive in every specification, though its magnitude varies.

We weight the gap regression by cell sample size rather than by inverse variance. Although inverse-variance weighting is standard in meta-analysis, it performs poorly here because bootstrap standard errors vary by orders of magnitude across cells.

One limitation of this approximation is that it assumes drivers are aware of the offers they receive. As documented in the main text, drivers sometimes ignore offers, even generous ones such as those in our Earnings Accelerator experiments. Suppose drivers see

Table A27: Robustness of IDB Elasticity Estimates

	(1) Non-Shifter	(2) Shifter	(3) Shifter – Non-Shifter	(4) Elasticities	(5) Driver-Periods
<i>Panel A: Polynomial degree</i>					
Baseline (degree 3)	0.602* (0.364)	1.100*** (0.188)	0.498 (0.416)	216	878,323
Degree 4	0.571* (0.317)	1.354*** (0.275)	0.784* (0.429)	206	864,708
<i>Panel B: Counterfactual window</i>					
Window = 100 trips	0.695** (0.333)	1.372*** (0.152)	0.677* (0.364)	219	911,662
<i>Panel C: Excluded region above threshold</i>					
Exclude 3 bins	0.859** (0.400)	1.399*** (0.210)	0.539 (0.443)	227	965,714
<i>Panel D: Trimming of extreme estimates</i>					
Trim $ e > 5$	0.956*** (0.216)	1.105*** (0.115)	0.149 (0.244)	199	805,636
Trim $ e > 25$	0.958** (0.479)	1.583*** (0.402)	0.625 (0.530)	231	953,124
<i>Panel E: Estimation method</i>					
Unweighted	0.419 (0.283)	1.062*** (0.177)	0.643** (0.327)	216	878,323
Median regression	0.450*** (0.034)	0.833*** (0.003)	0.382*** (0.034)	216	878,323

Note: Each row reports estimates from a separate regression of cell-level trip elasticities on a shifter indicator, with threshold fixed effects. The baseline specification uses a polynomial of degree 3, a counterfactual window of 50 trips, excludes 4 bins above the threshold, trims estimates with $|e| > 10$, and weights by cell sample size. Each panel varies one parameter from this baseline. Robust standard errors are in parentheses. See Appendix E.2 for details on the bunching estimation procedure. Levels of significance: *10%, ** 5%, and *** 1%.

the IDB offer only with probability α . In this case, the scaled excess mass approximates $\alpha \times \Delta z^{x,y}$ rather than $\Delta z^{x,y}$.

E.4 Calculating Standard Errors

We estimate standard errors for elasticities and bunching estimates using the following procedure:

1. Fit the pooled observed trip density—aggregating across all bonus amounts and groups—to a polynomial in distance from the threshold:

$$p_z^x = \sum_{d=0}^D \beta_d (z - x)^d + \sum_{j=x-z_L}^{x+z_U} \gamma_j 1\{z = j\} + \xi_z \quad (9)$$

using a choice of polynomial degree D , upper window z_U , and lower window z_L . The excluded region thus spans $[x - z_L, x + z_U]$: the “Excluded Bins” parameter reported in Table 6 is the number of bins above the threshold (z_U), while the threshold-specific number of bins below (z_L) is listed in the last column of Appendix Table A25. In our baseline estimates, we pool across bonus amounts and shifter and non-shifter groups within each threshold x and activity stratum, so the counterfactual density is common to all cells at a given threshold. In some robustness exercises, we estimate Equation (9) separately by group.

2. Calculate the excess mass for drivers in group g (defined by shifter status and bonus amount y) at threshold x . We subtract the pooled predicted density from the group-specific observed density in the excluded region $[x, x + z_U]$:

$$B_{x,y,g} = \sum_{z=x}^{x+z_U} \left(p_{z,y,g}^x - \sum_{d=0}^D \hat{\beta}_d (z - x)^d \right)$$

3. Calculate the scaled excess mass by dividing by the predicted counterfactual density at the threshold, $\hat{h}_0^x = \hat{\beta}_0$ (the regression constant, since the polynomial is centered

at x):

$$\hat{M}_{x,y,g} = B_{x,y,g} / \hat{h}_0^x$$

4. Estimate the elasticity using the structural notch approximation (Section E.2):

$$\hat{e}_{x,y,g} = \frac{1}{2} \cdot \frac{\left(\hat{M}_{x,y,g}\right)^2}{(x - \hat{M}_{x,y,g}) \cdot (y/w)}$$

where w is median earnings per trip, y is the bonus amount, and $\hat{M}_{x,y,g} \approx \Delta z_{x,y,g}$ is the implied shift in the trip distribution (Kleven, 2016).

5. For each of 500 bootstrap replications:
 - (a) Resample with replacement from the residuals of Equation (9) to construct a bootstrapped pooled density.
 - (b) Re-estimate the polynomial on the bootstrapped density.
 - (c) Re-calculate $\hat{M}_{x,y,g}$ using the group-specific observed density and the new predicted counterfactual.
 - (d) Re-calculate $\hat{e}_{x,y,g}$.

We resample the counterfactual polynomial residuals but hold the group-specific bin counts fixed. As a result the bootstrap standard errors capture uncertainty in the polynomial fit but not sampling variation in the group-specific counts. We address this by using cell-size weighting rather than inverse-variance weighting in the gap regressions (see Section E.2).

6. The bootstrap standard error is the standard deviation of the elasticity estimates across bootstrap replications.